



Enclustra User Schematics

Mercury PE1 Revision 4.6

Disclaimer

User schematics do not include proprietary Enclustra design elements, such as power supply circuits. These circuits use optimized designs which are extremely compact and remain proprietary to Enclustra. The user manual provides all necessary information to use the module and its interfaces.

User schematics are provided after purchase of Enclustra hardware, and may also be provided in certain cases before purchase, to assist in product evaluation.

No part of the schematics may be copied or modified without written permission from Enclustra.

Full schematics, including a full bill-of-materials, may be available through the purchase of a hardware licence for the product in question.

Please contact Enclustra Sales for more information.

Note: DNE = Do Not Equip (parts not equipped by default)

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Mercury PE1 Base Board

Revision 4.6

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- Sheet 2: Mercury Module Connectors A/B
- Sheet 3: Mercury Module Connector C
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Pin connection diagram for the FX10A-168P-SV module. The diagram shows two rows of pins, labeled B1 through B168. The left side lists various input/output signals such as IOE D0 LED0#, MGT TX0 P, IOC D0 P, FMC CLK1 M2C P, IOD D0 P, IOD D4 P, FMC LA33 P, FMC LA32 P, FMC LA31 P, FMC LA30 P, FMC LA29 P, FMC LA28 P, FMC LA27 P, FMC LA26 P, FMC LA25 P, FMC LA18 CC P, FMC LA24 P, FMC LA23 P, FMC LA22 P, FMC LA21 P, FMC LA17 CC P, FMC LA20 P, FMC LA19 P, and VMON B167. The right side lists corresponding signals for the other row, including MGT REFCLK0 P, VMON B8, MGT REFCLK1 P, MGT RX0 P, MGT RX1 P, MGT RX2 P, MGT RX3 P, IOC D2 P, IOC D3 N, IOC D6 P, IOD D2 P, IOD D3 N, IOD D6 P, IOD D7 N, FMC LA16 P, FMC LA16 N, FMC CLK0 M2C P, FMC CLK0 M2C N, FMC LA15 P, FMC LA15 N, FMC LA14 P, FMC LA14 N, FMC LA13 P, FMC LA13 N, FMC LA12 P, FMC LA12 N, FMC LA11 P, FMC LA11 N, FMC LA10 P, FMC LA10 N, FMC LA09 P, FMC LA09 N, FMC LA08 P, FMC LA08 N, FMC LA07 P, FMC LA07 N, FMC LA06 P, FMC LA06 N, FMC LA05 P, FMC LA05 N, FMC LA04 P, FMC LA04 N, FMC LA00 CC P, FMC LA00 CC N, FMC LA03 P, FMC LA03 N, FMC LA02 P, FMC LA02 N, FMC LA01 P, FMC LA01 N, and VMON B168. The diagram also shows power supply connections for VCC_OUT_B, VCC_IO_B, and VCC_3V3_MOD.

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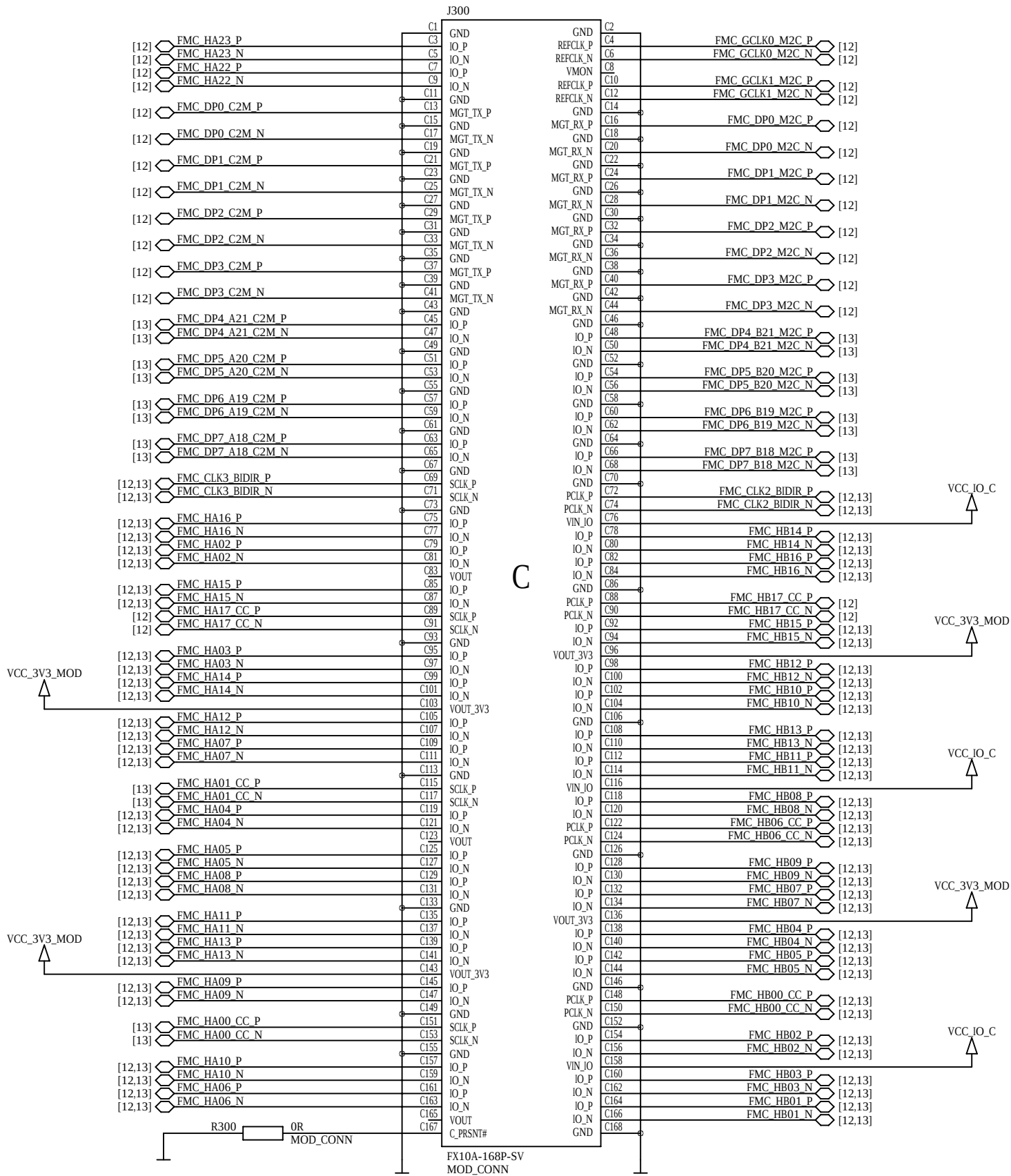
B

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A

Mercury Module Connector C

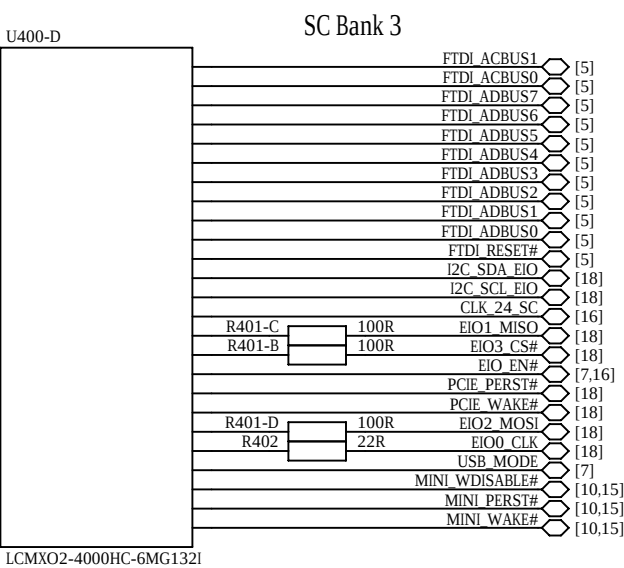
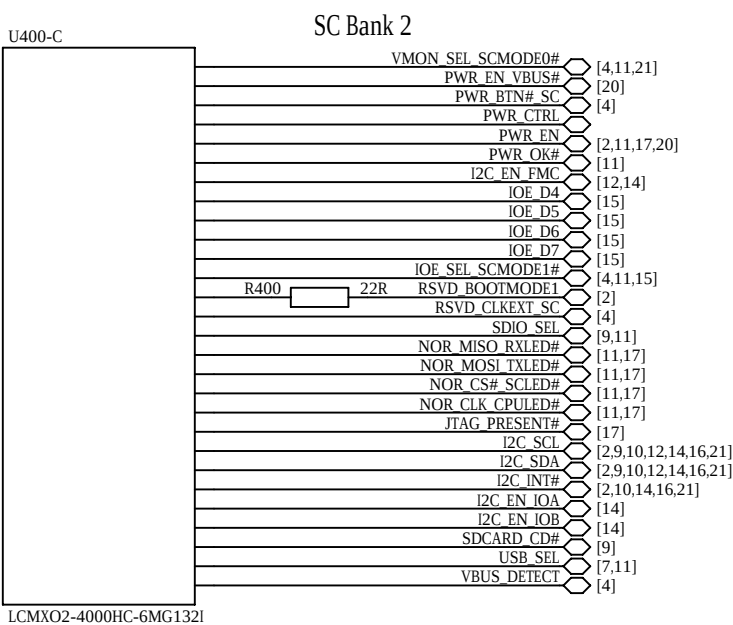
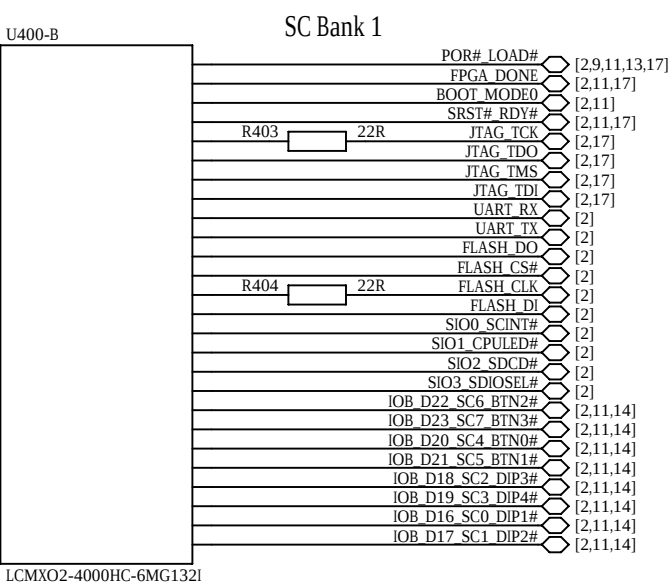
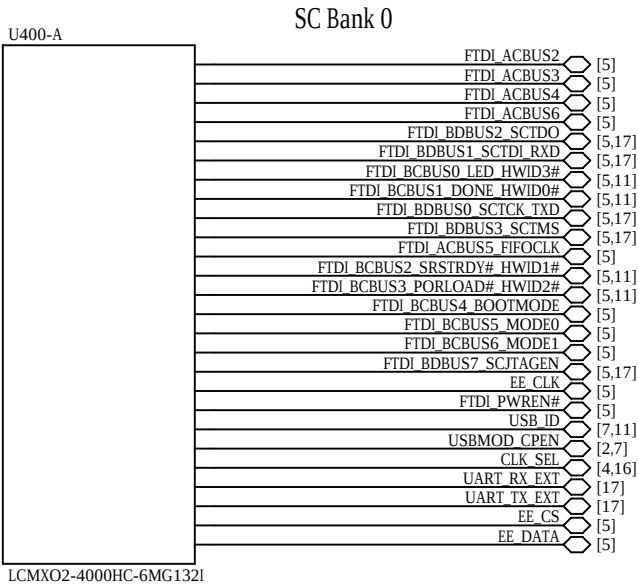


System Controller PLD

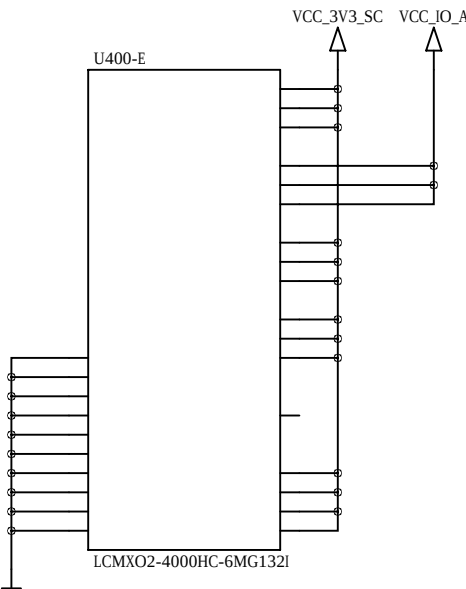
The pinout is not included in the user schematics.

Programming the system controller PLD with a user bitstream voids the board warranty.

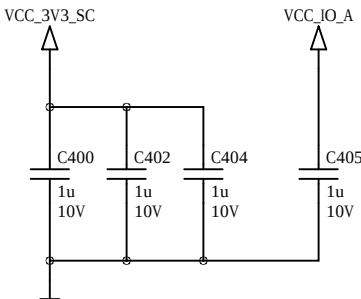
I2C Address: 0 0 0 1 1 0 1 Rw



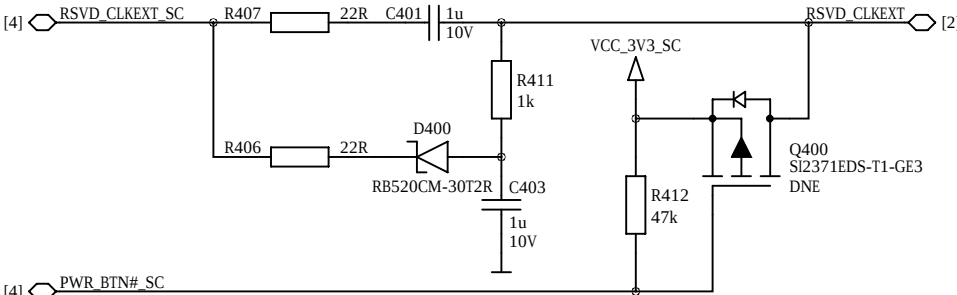
SC Power



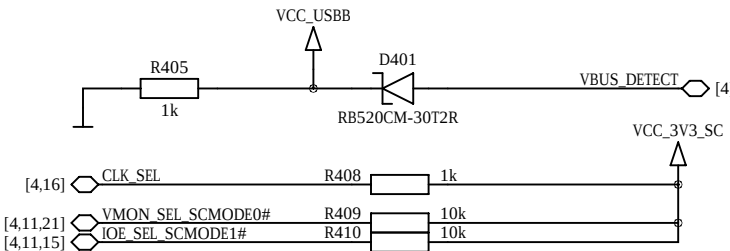
SC Decoupling



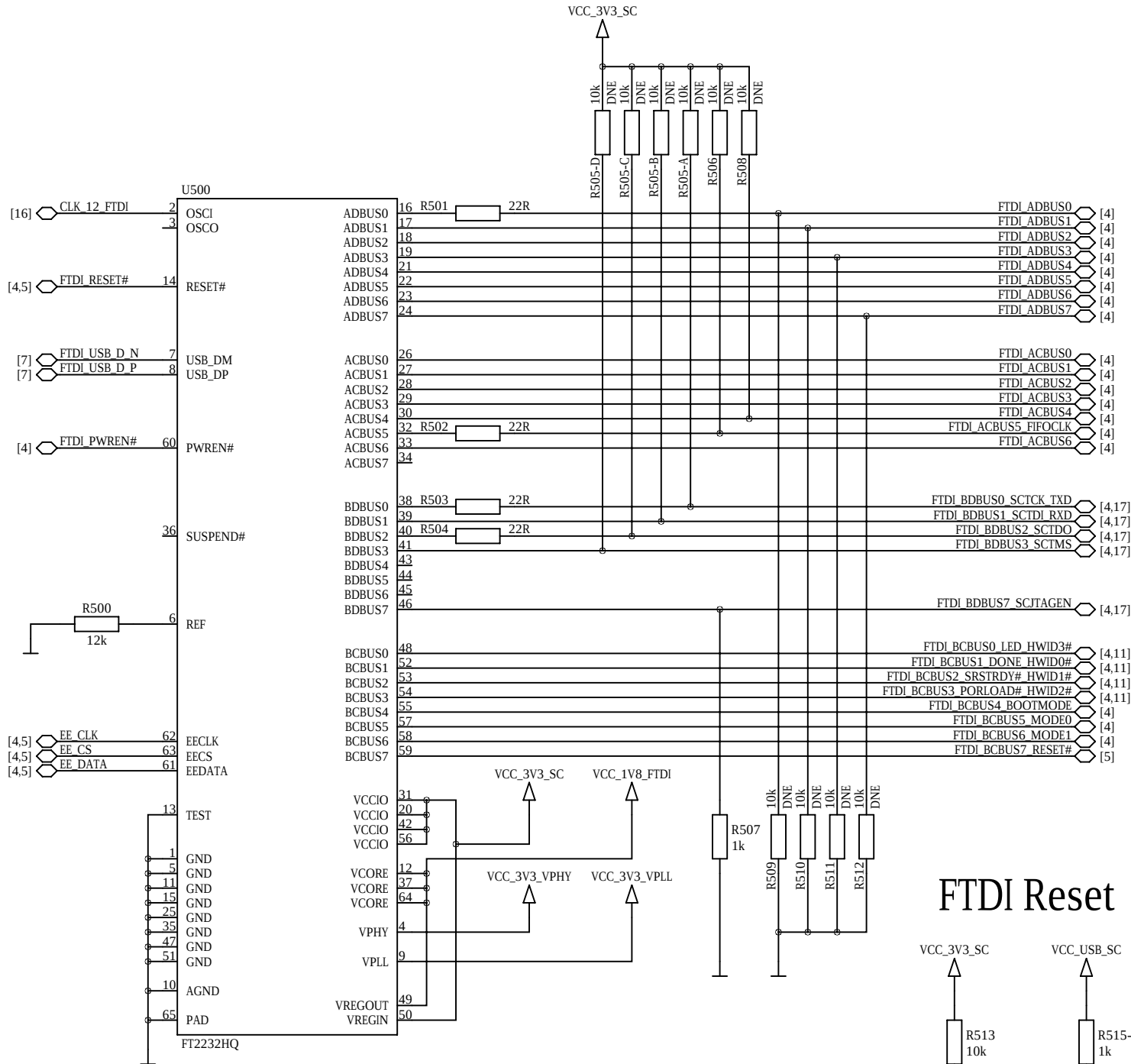
Synchronous Module Clock



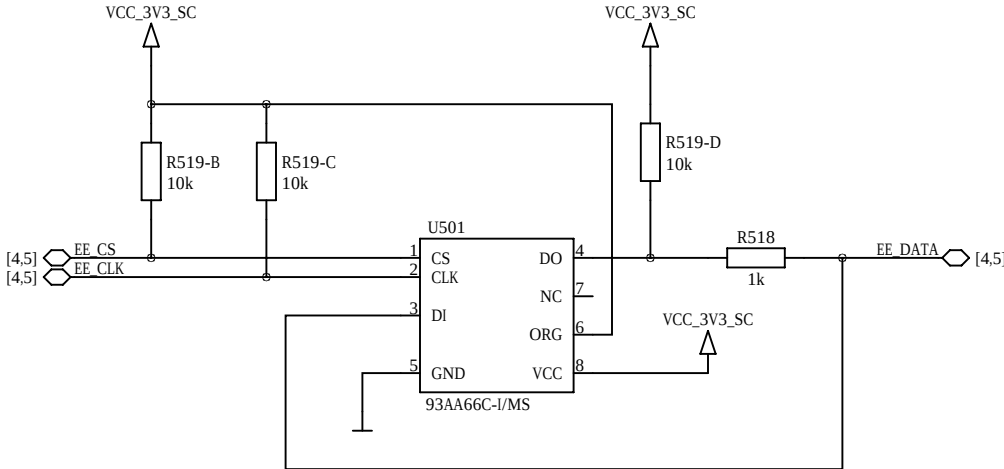
USB VBUS Detection



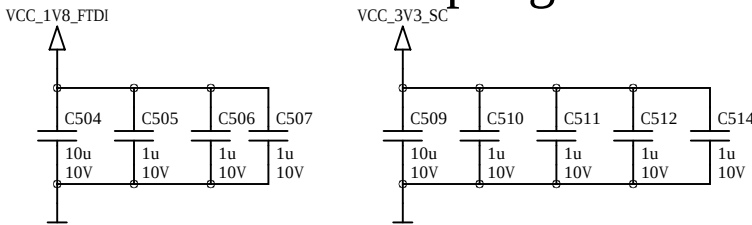
FTDI USB 2.0 Device Controller



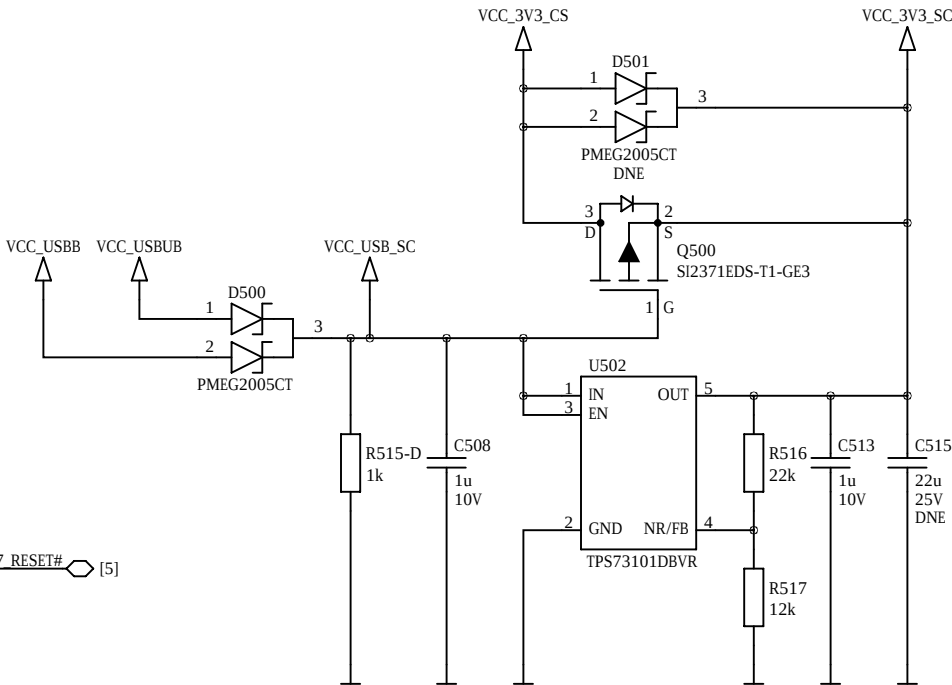
FTDI EEPROM



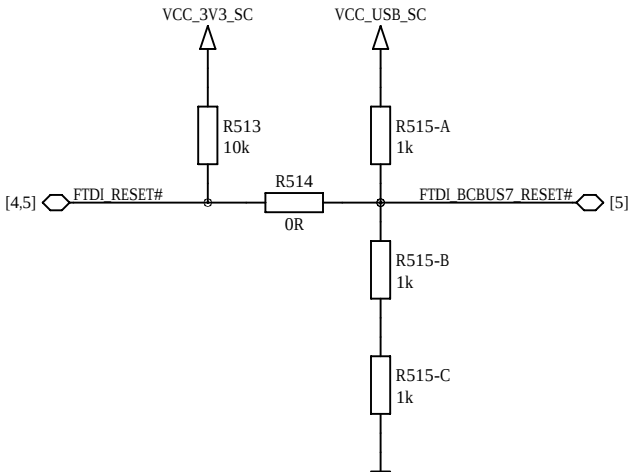
FTDI Decoupling



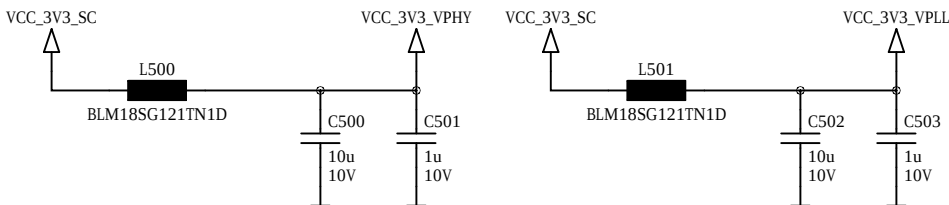
LDO 5V to 3.3V 0.15A



FTDI Reset

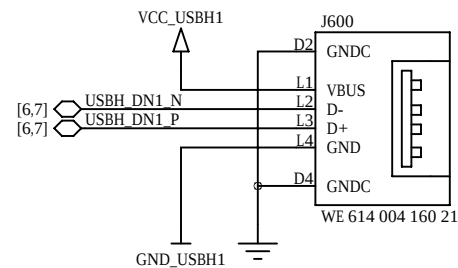


FTDI Power Filters



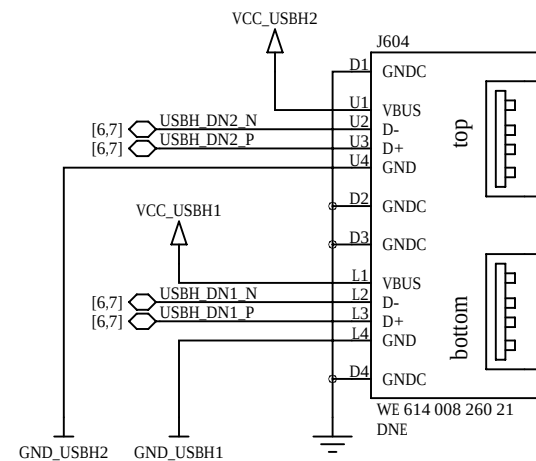
USB 2.0 USB Host

right-angle, single



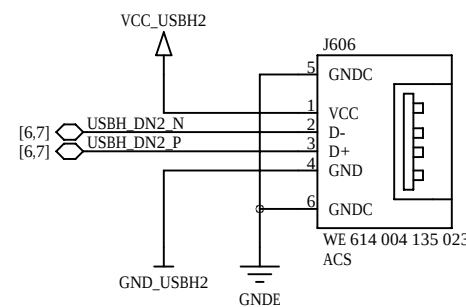
USB 2.0 USB Host

right-angle, dual, optional

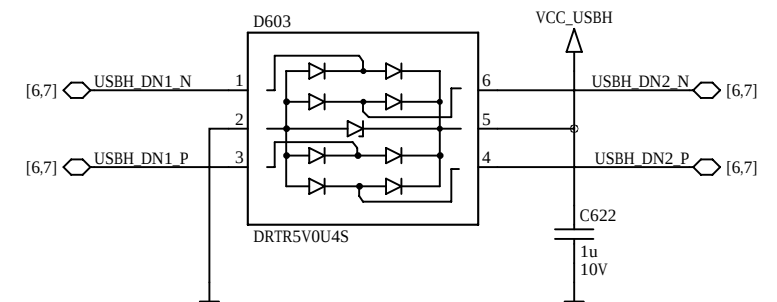


USB 2.0 USB Host

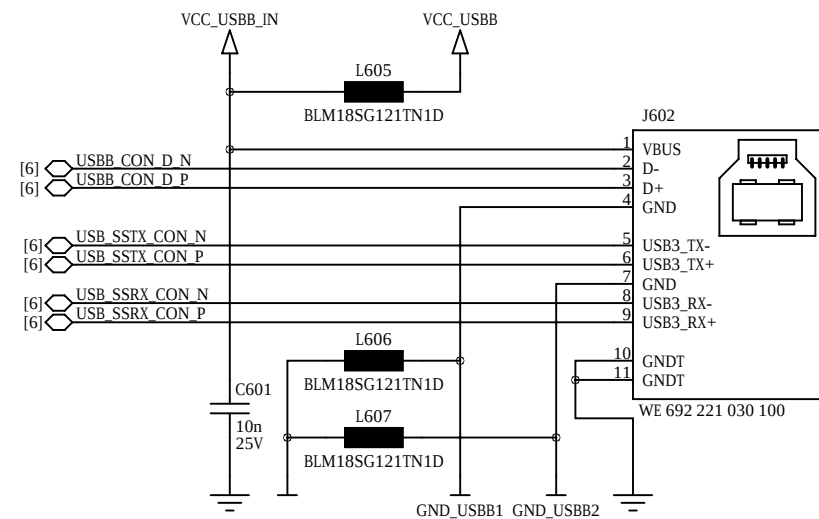
vertical



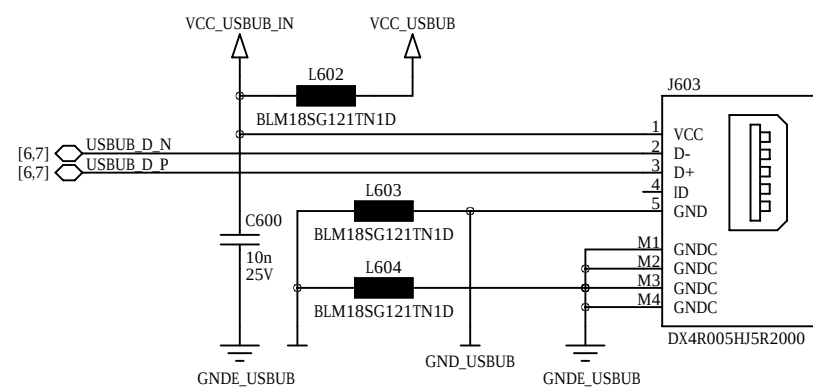
USB 2.0 Host ESD Protection



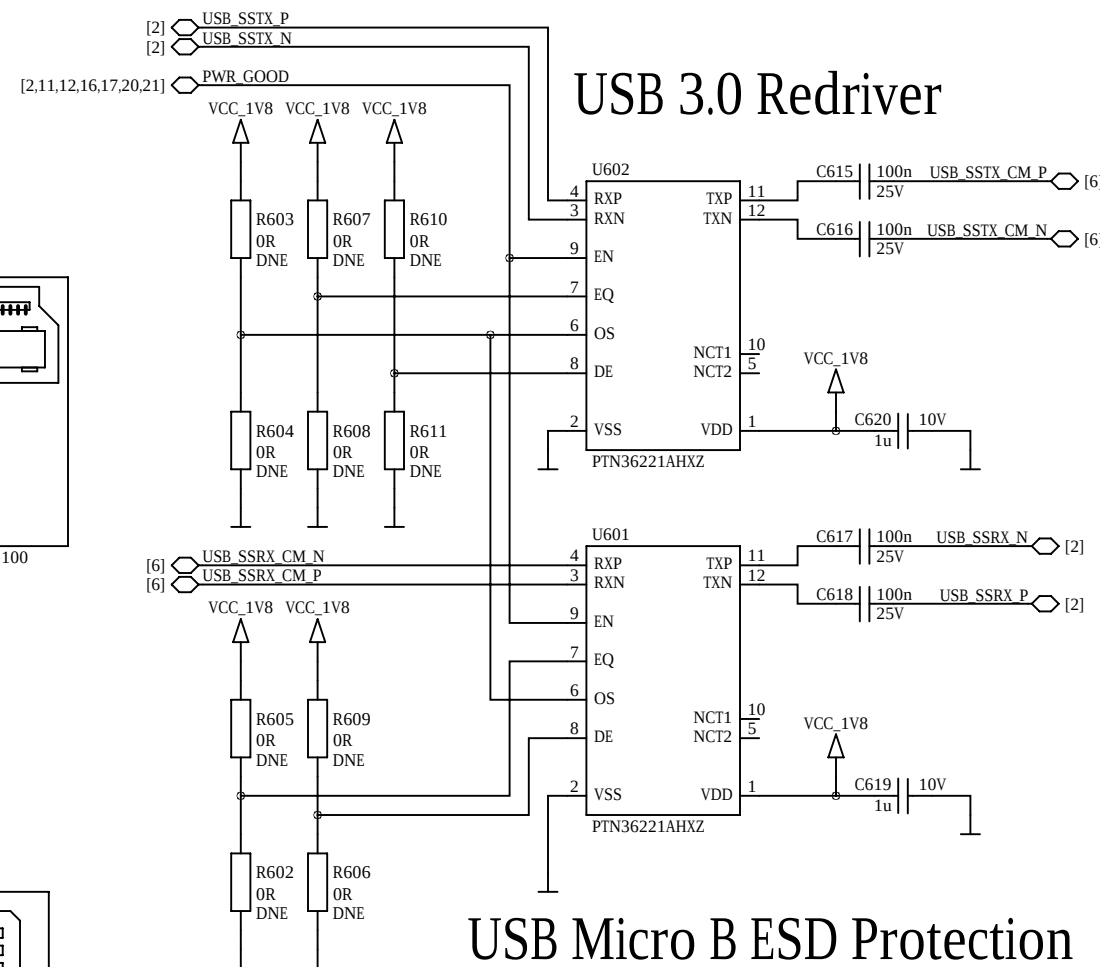
USB 3.0 Device Connector



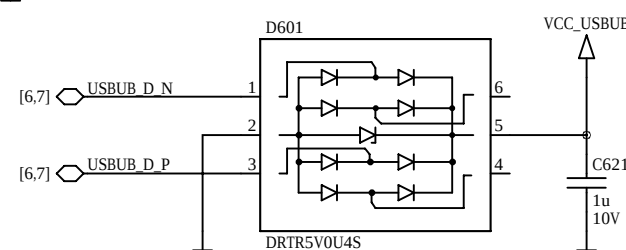
USB Micro B Connector



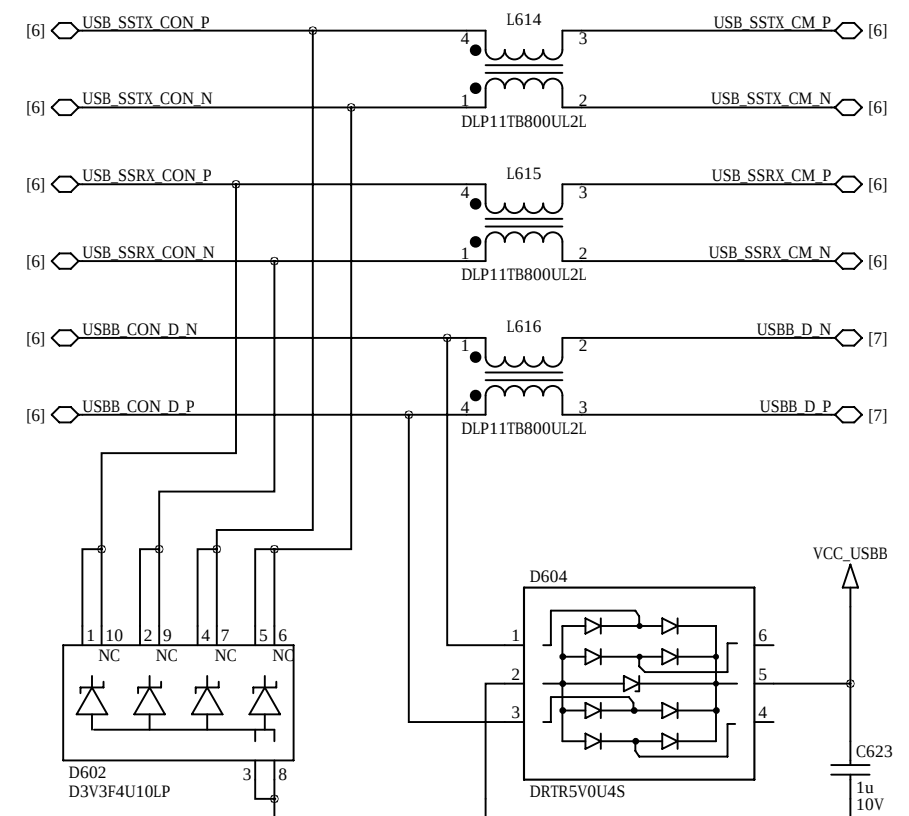
USB 3.0 Redriver



USB Micro B ESD Protection



USB 3.0 Device ESD Protection



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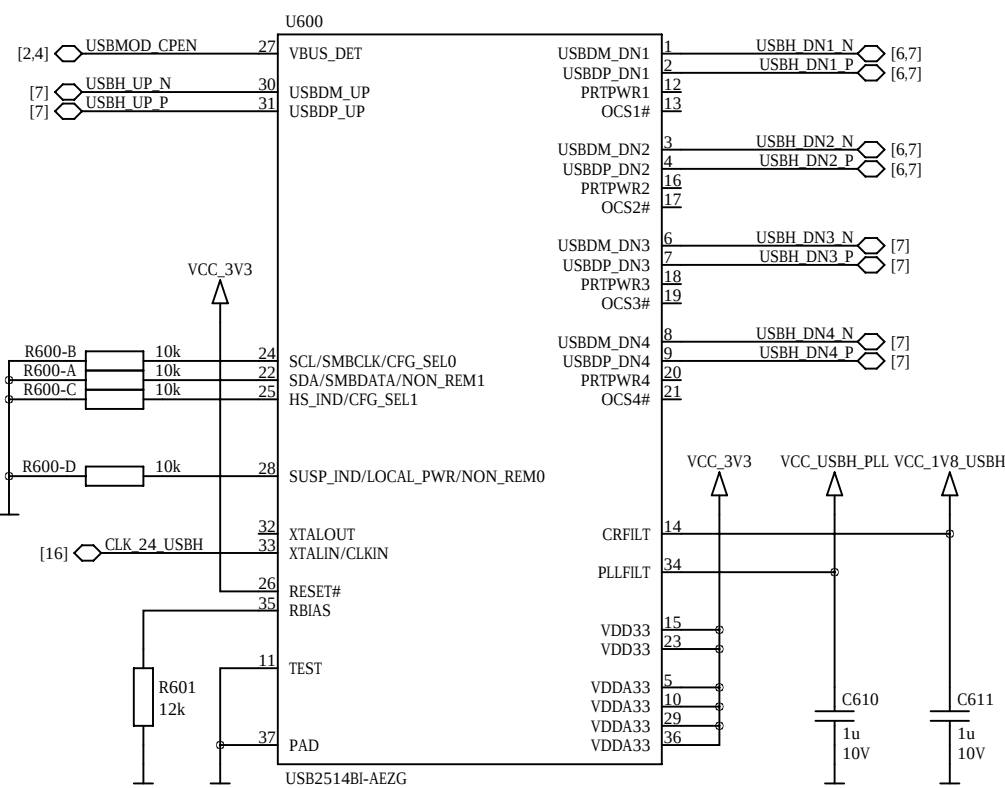
D

C

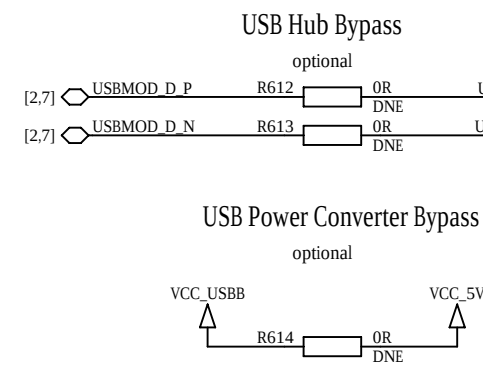
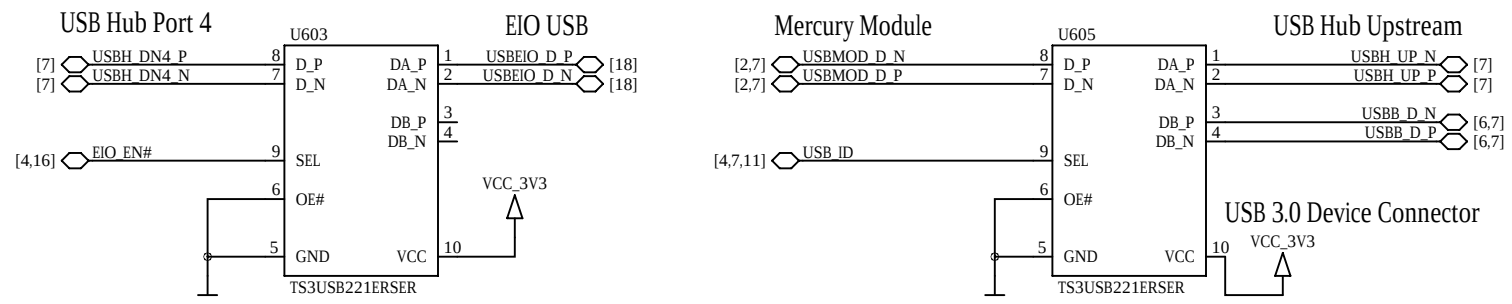
B

A

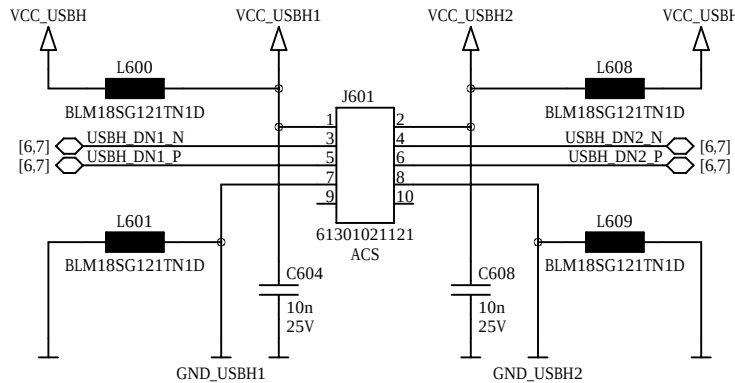
USB 2.0 Hub



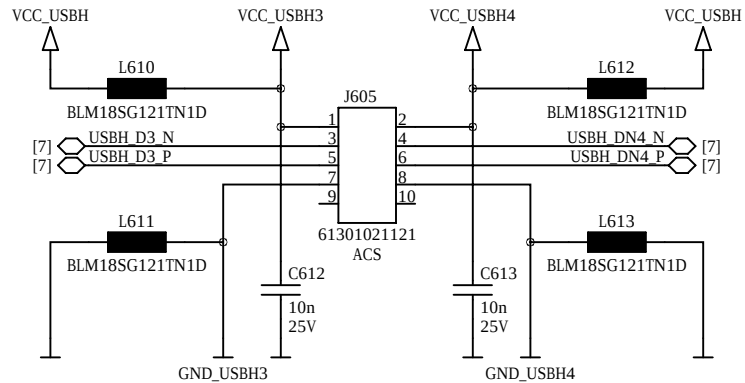
USB Multiplexers



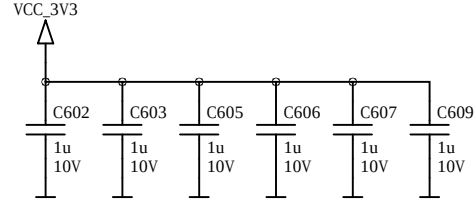
USB Header Port 1/2



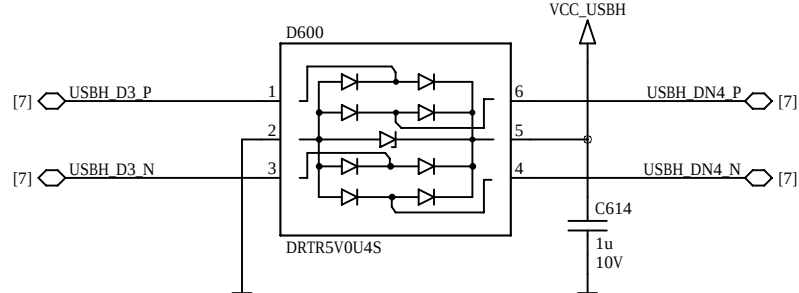
USB Header Port 3/4



USB 2.0 Hub Decoupling

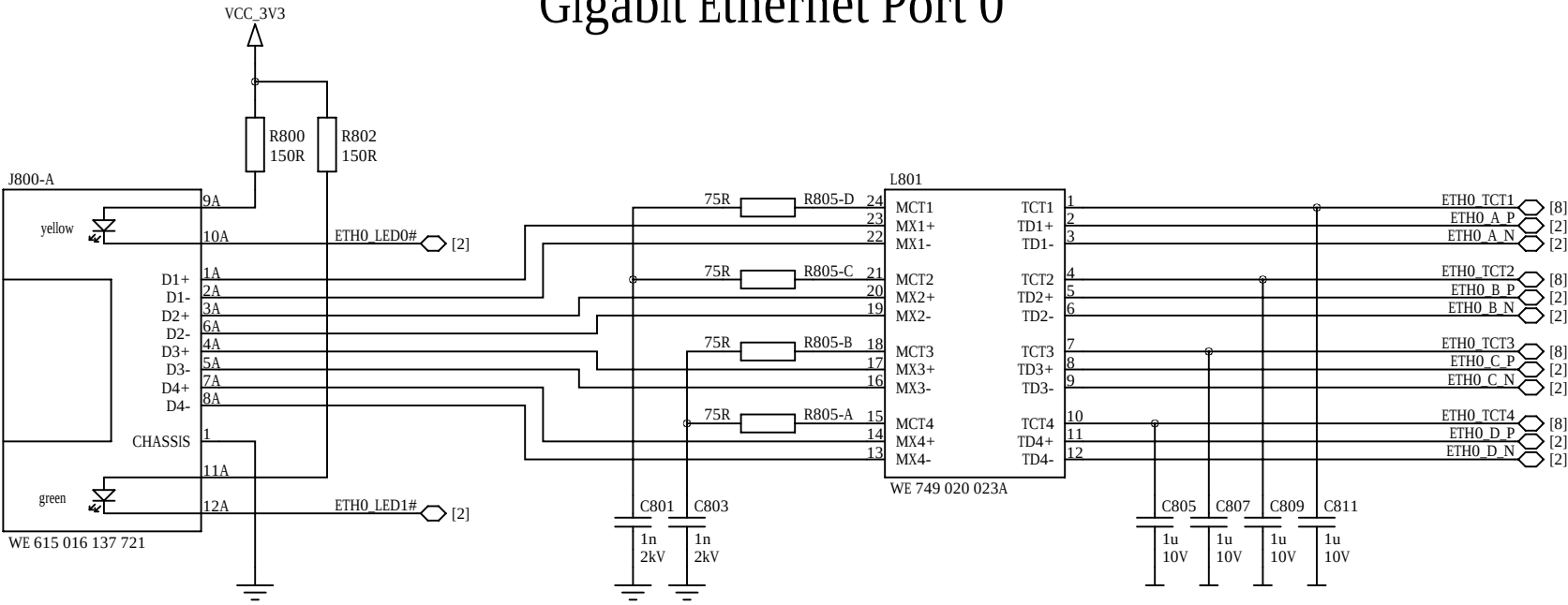


USB Header ESD Protection

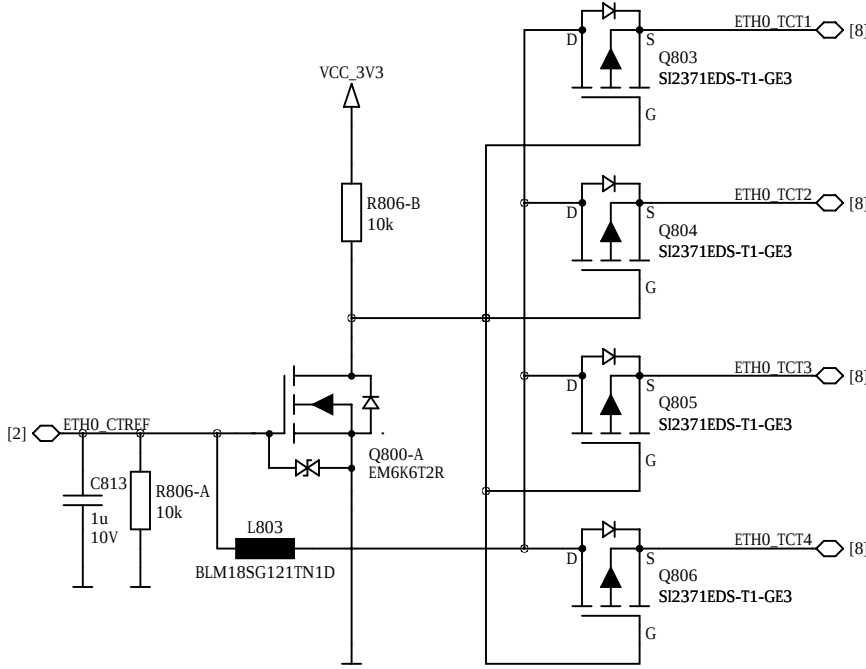


Copyright	© 2020 by Enclustra GmbH	Sheet Name	7_USB_HUB_MULTIPLEXERS	Customer No	0000	Revision	R4.6	DNE = do not equip	Confidential
Company	Enclustra FPGA Solution Center	Project	Mercury PE1	Project No	423	Designed	MHEI	Date	13 Apr 2021
								Sheet/sheets	7 / 23

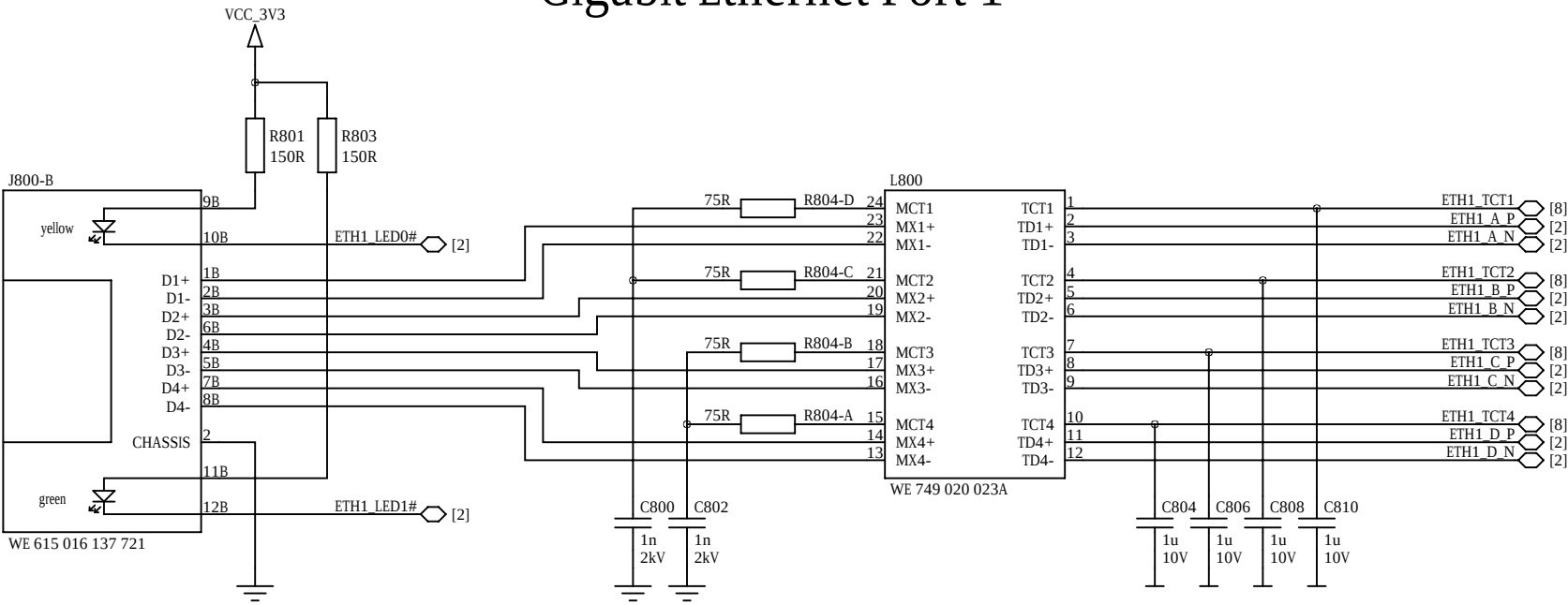
Gigabit Ethernet Port 0



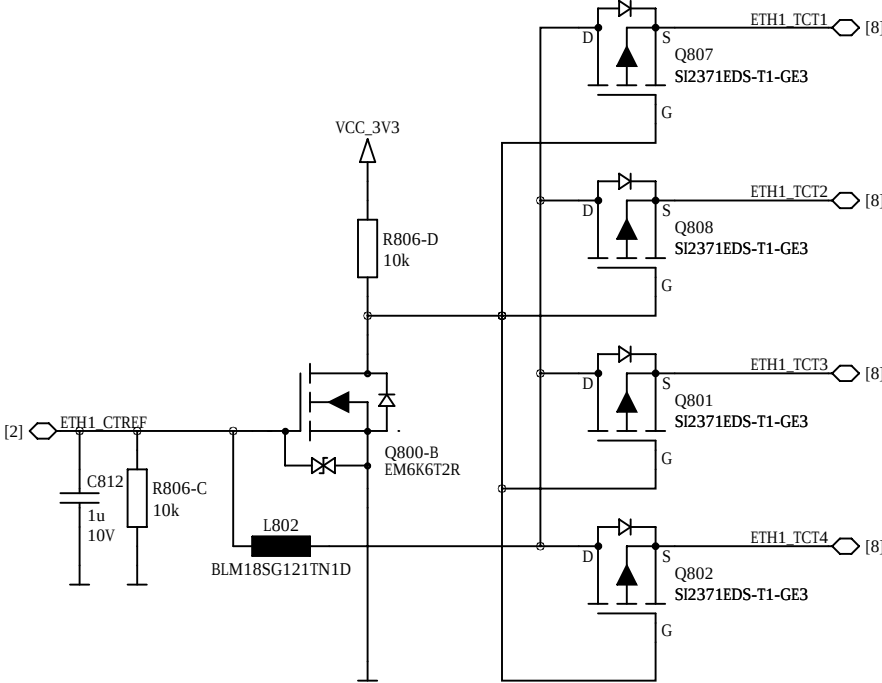
Magnetics Center Tap Port 0



Gigabit Ethernet Port 1



Magnetics Center Tap Port 1



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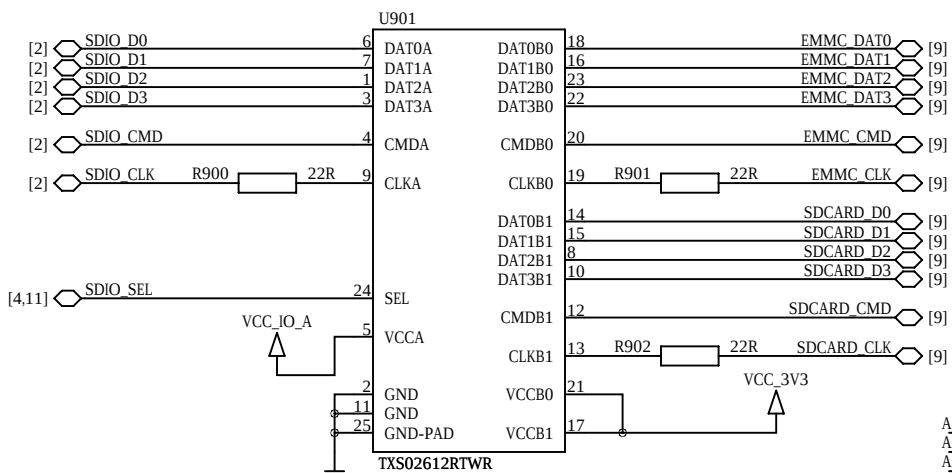
B

B

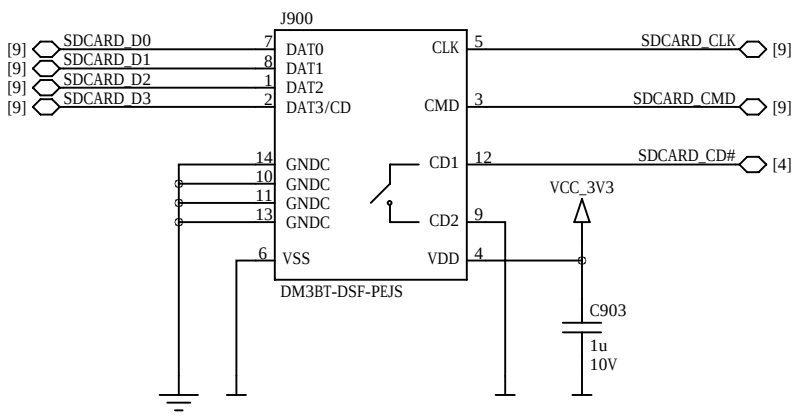
A

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SDIO Level Shifter

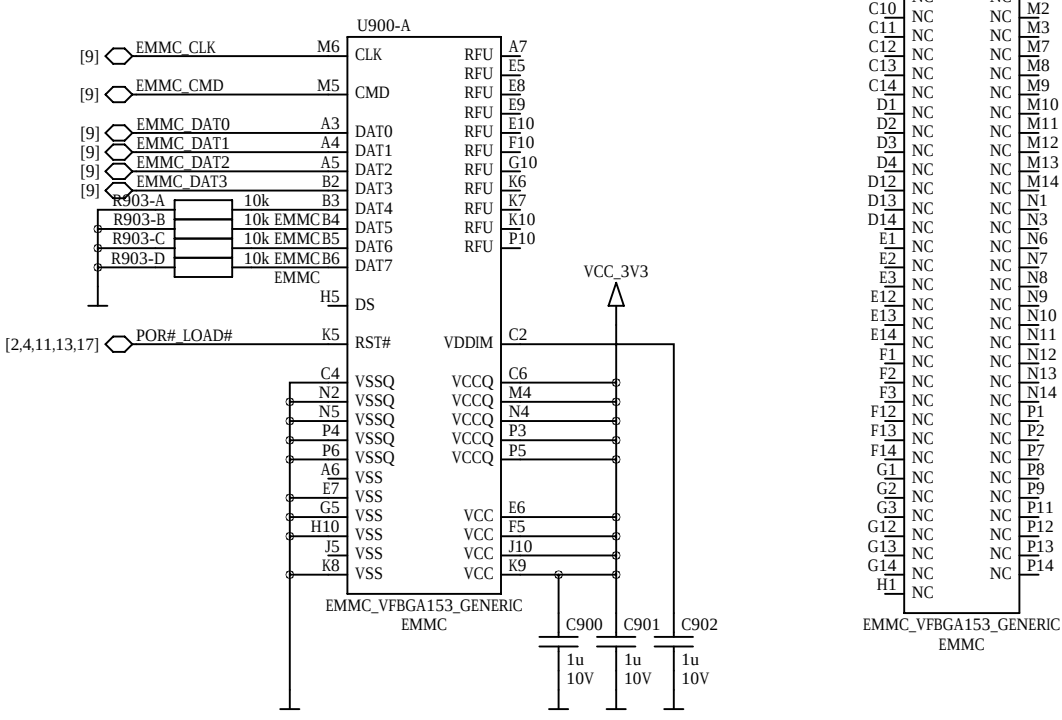


Micro SD-Card Slot

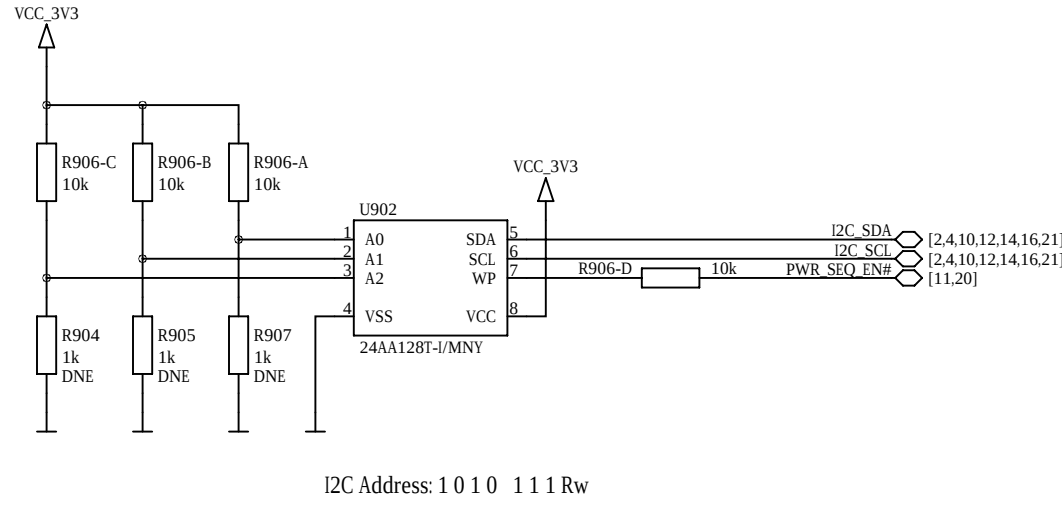


eMMC Managed NAND Flash

optional in some configurations



User EEPROM

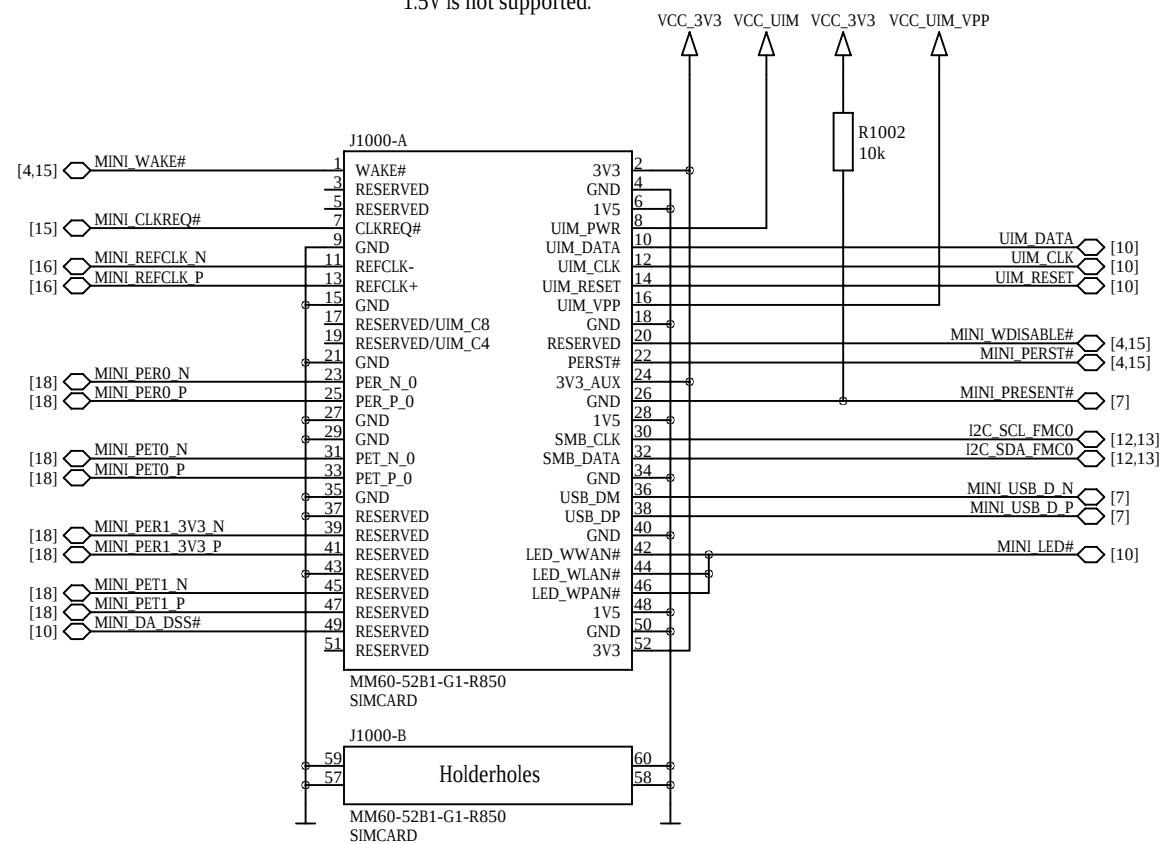


PCIe Mini / Mini SATA Card Holder

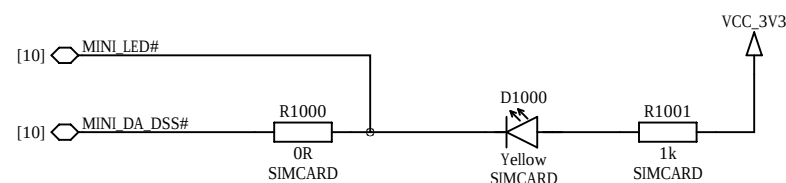
optional in some configurations

By default USB only. PCIe and SATA optional.

1.5V is not supported.

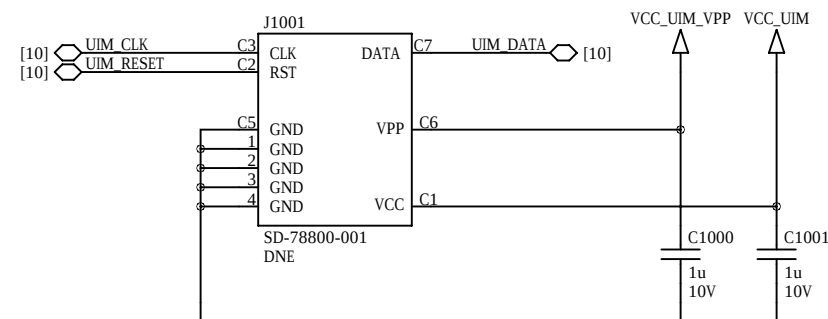


Mini Card Activity LED



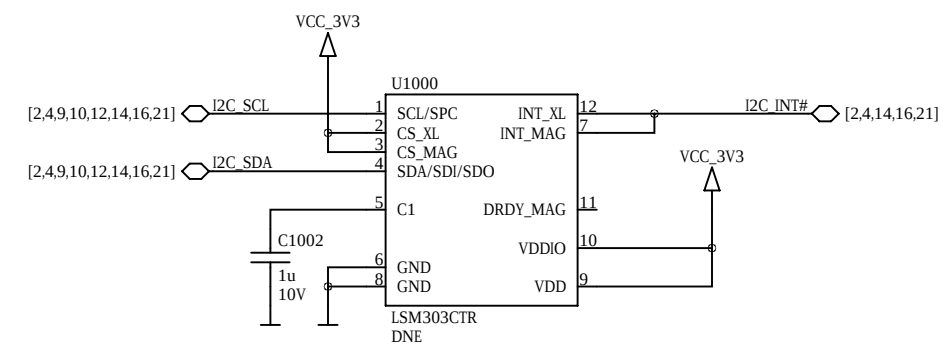
SIM Smart Card Slot

optional



Accelerometer / Magnetometer / Temperature Sensor

optional in some configurations



I2C Addresses

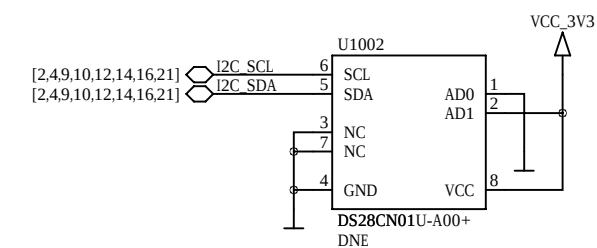
Accelerometer: 0 0 1 1 1 0 1 Rw

Magnetometer: 0 0 1 1 1 1 0 Rw

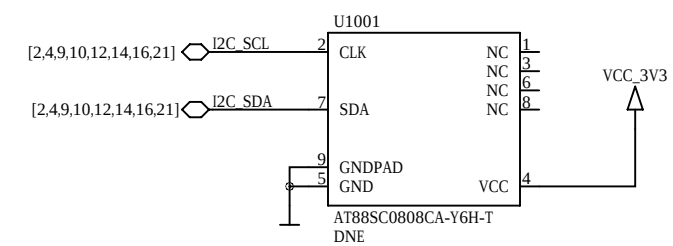
Secure EEPROMs

optional

for Enclustra use only

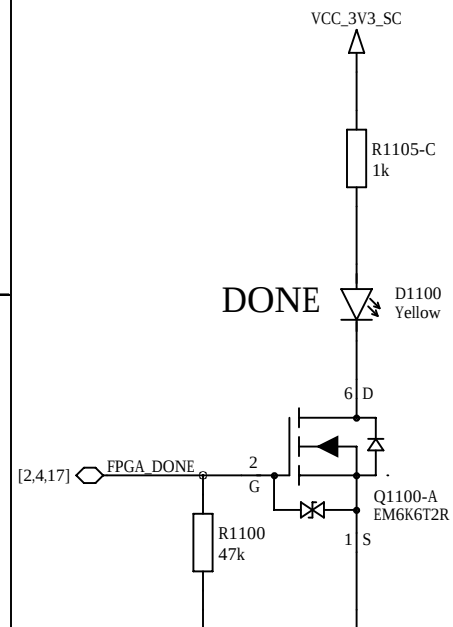


I2C Address: 1 0 1 0 1 0 0 Rw

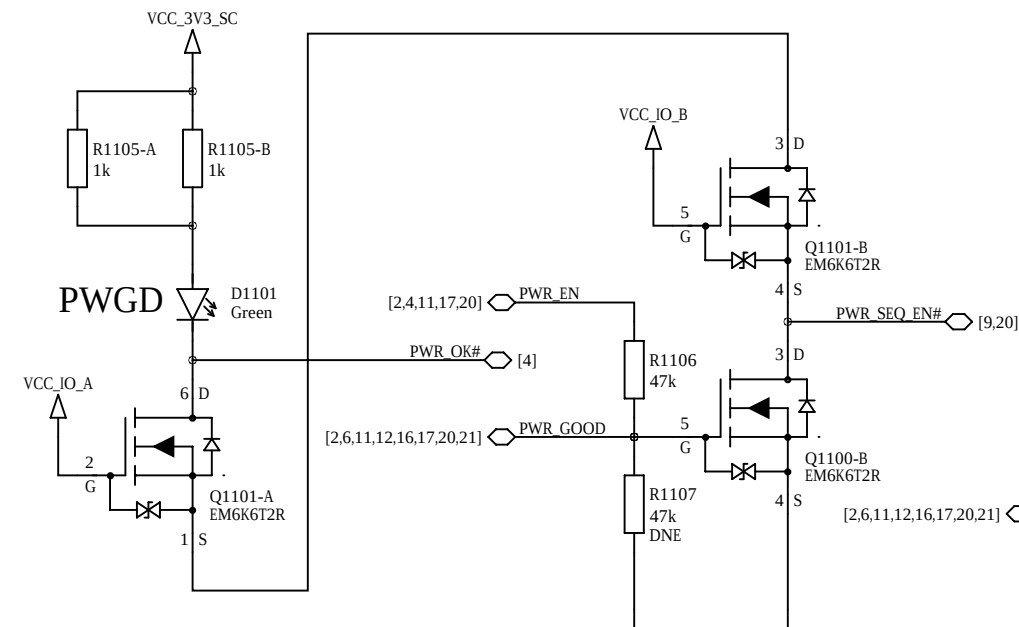


I2C Address: 0 0 0 1 X 1 1 Rw

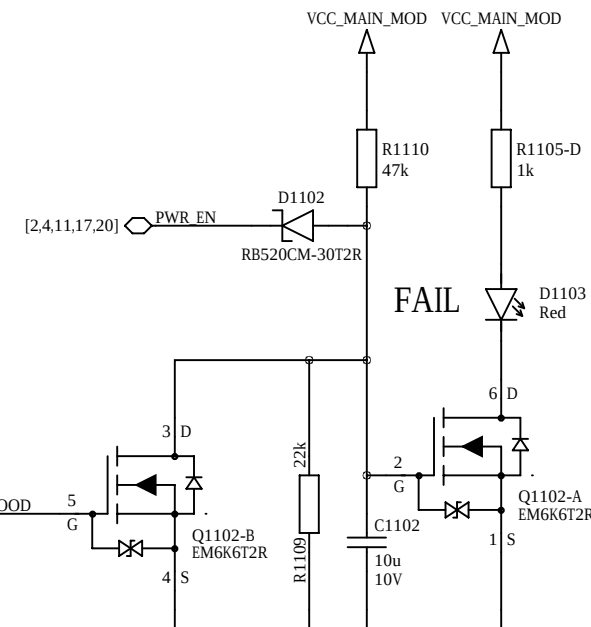
Done LED



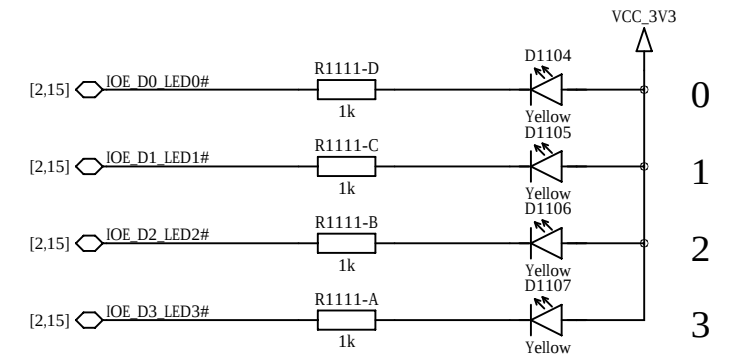
Power Good LED



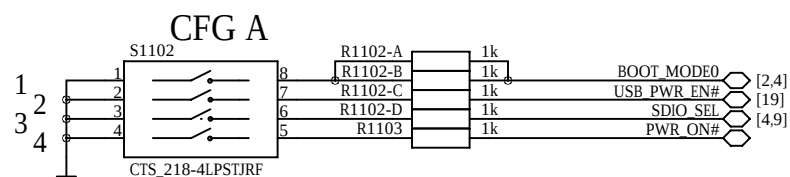
Power Fail LED



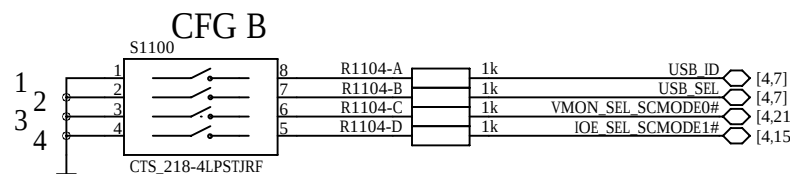
User LEDs



Board Configuration

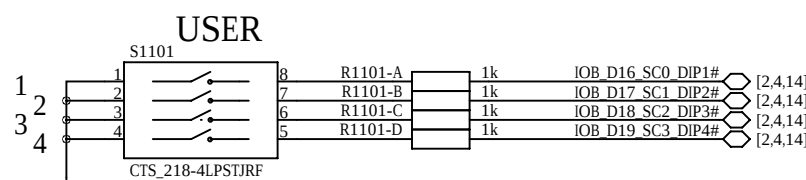


Default: 1-4 open



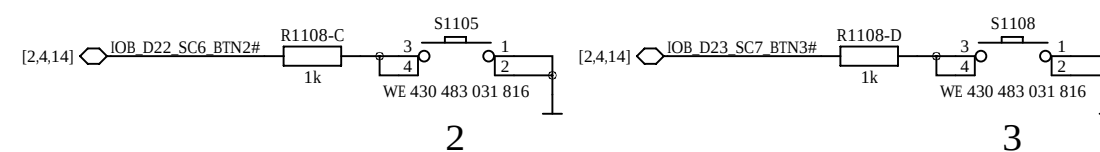
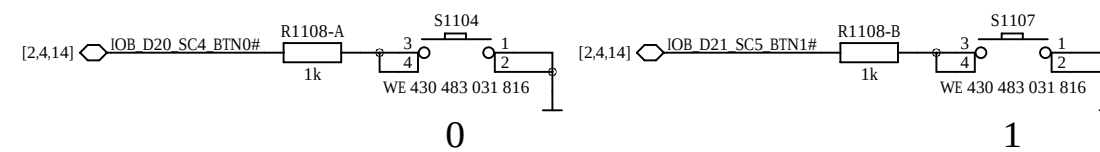
Default: 1-4 open

User DIP Switches

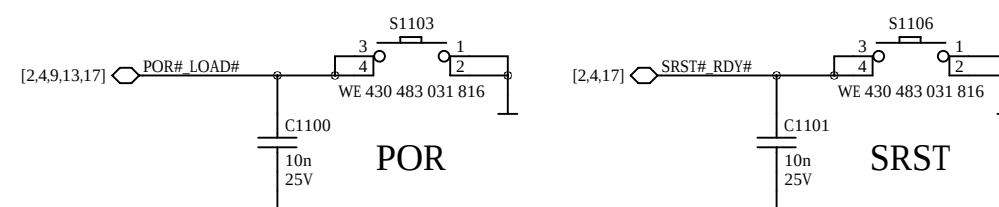


Default: 1-4 open

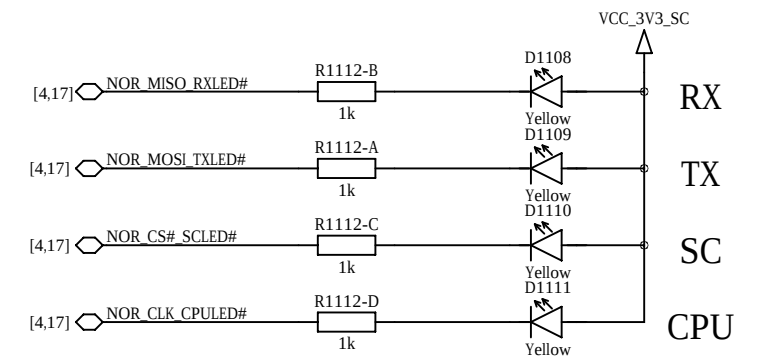
User Buttons



Reset Buttons

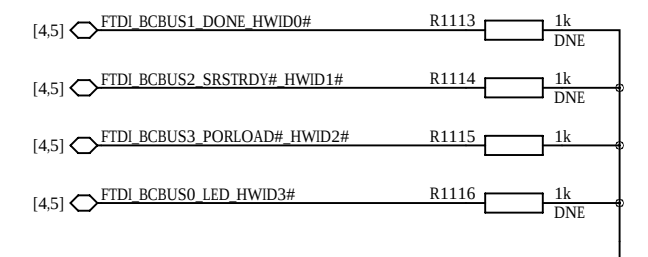


System Controller LEDs



Hardware Revision

HWID = 1 0 1 1



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FMC HPC/LPC Connector 0

FMC I2C Bus

D

C

B

A

D

C

B

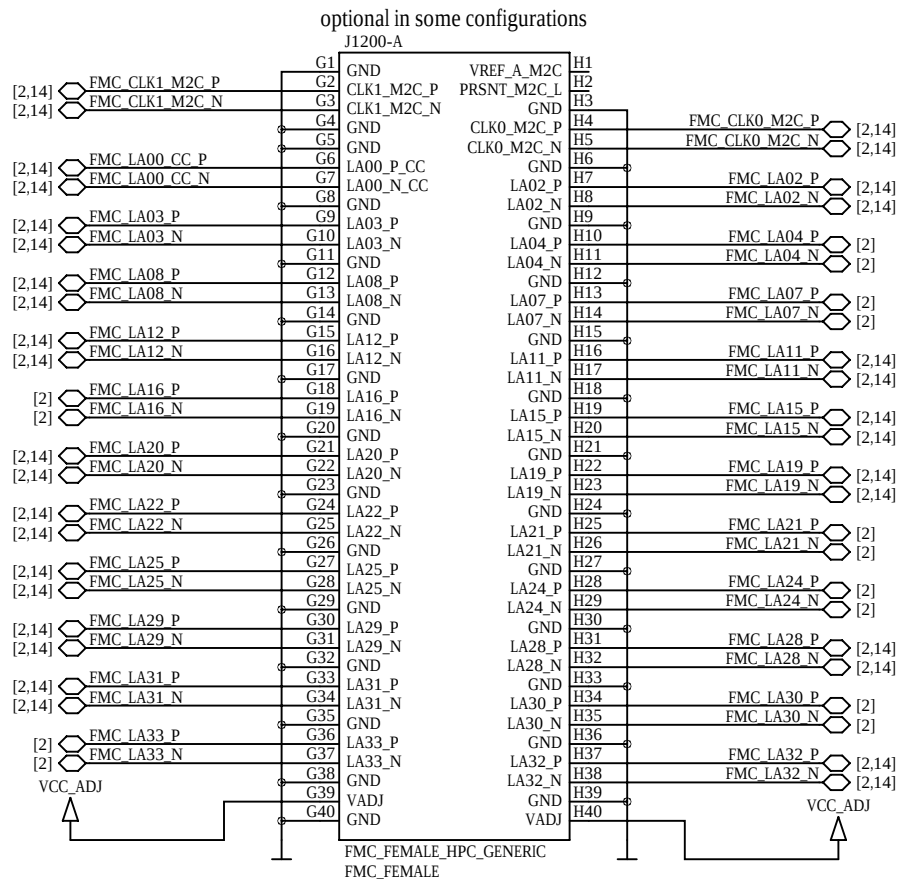
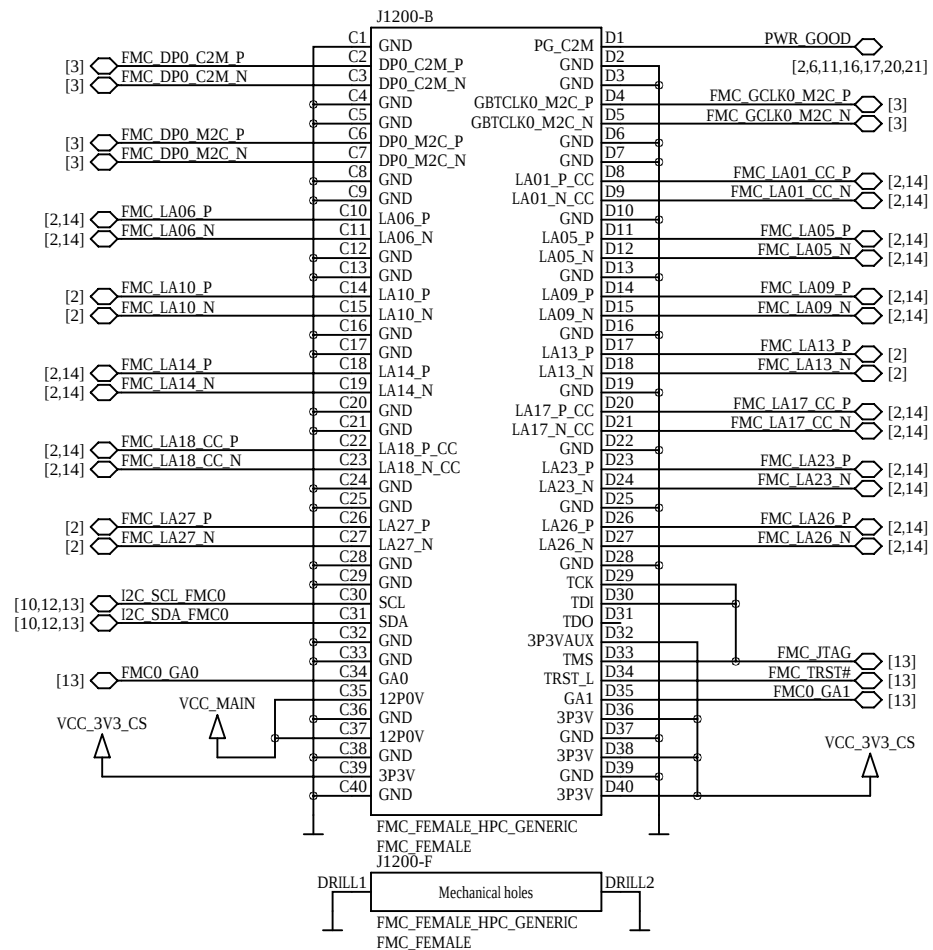
A



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Company Enclustra FPGA Solution Center

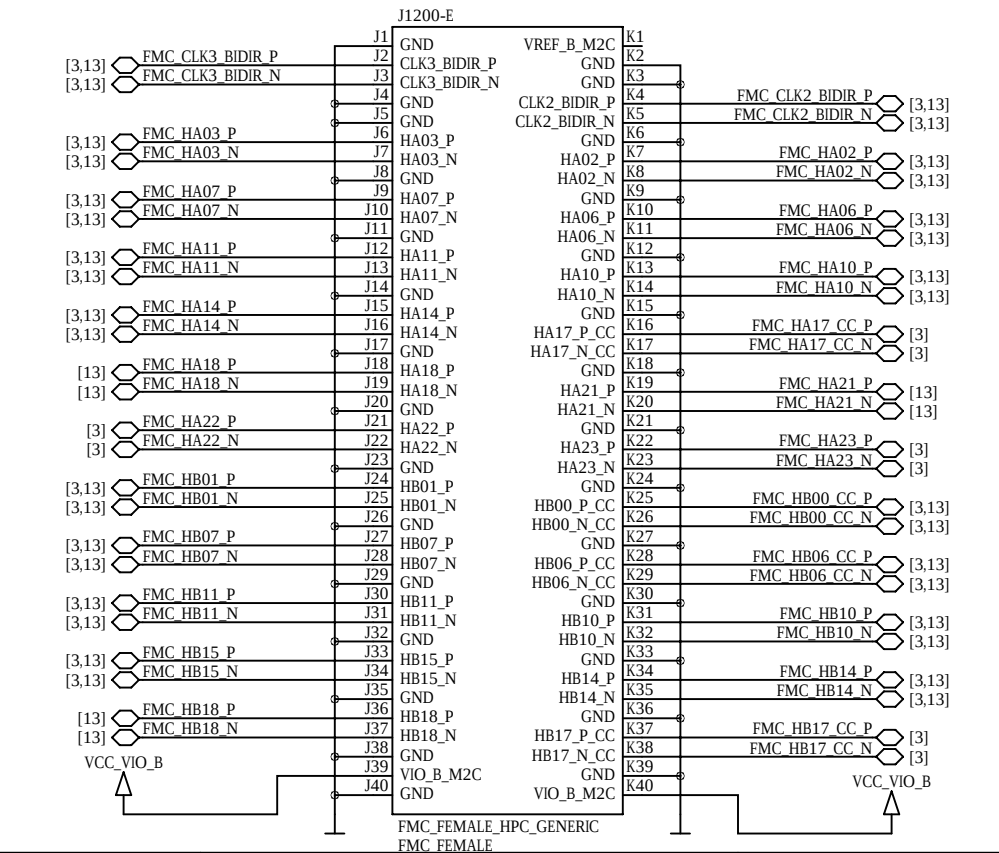
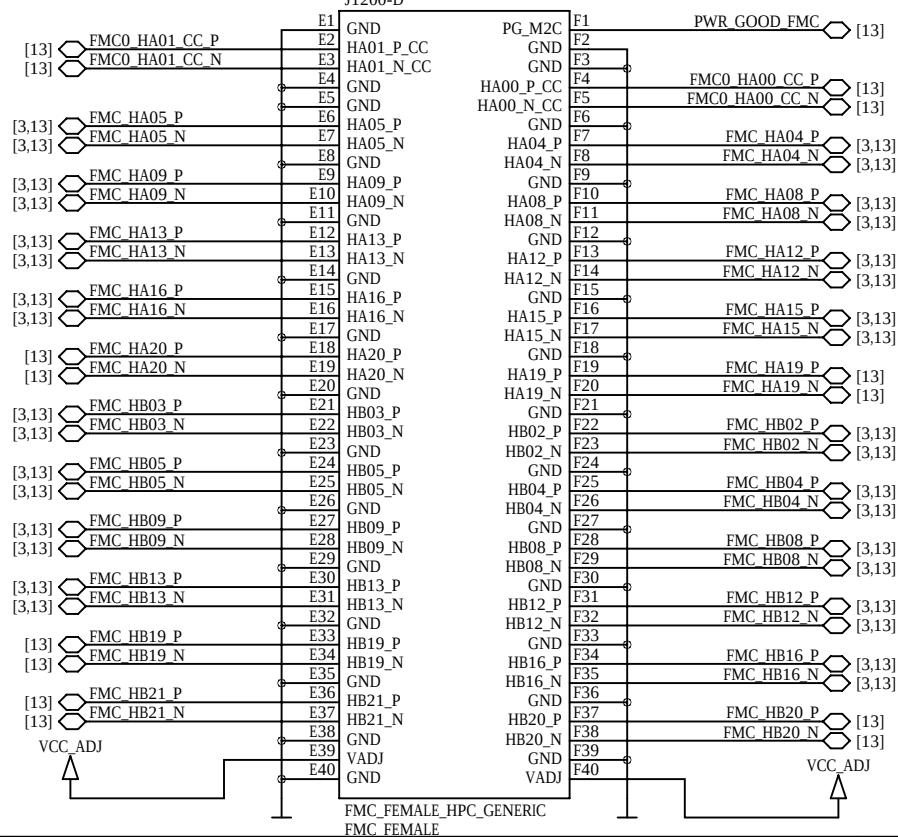
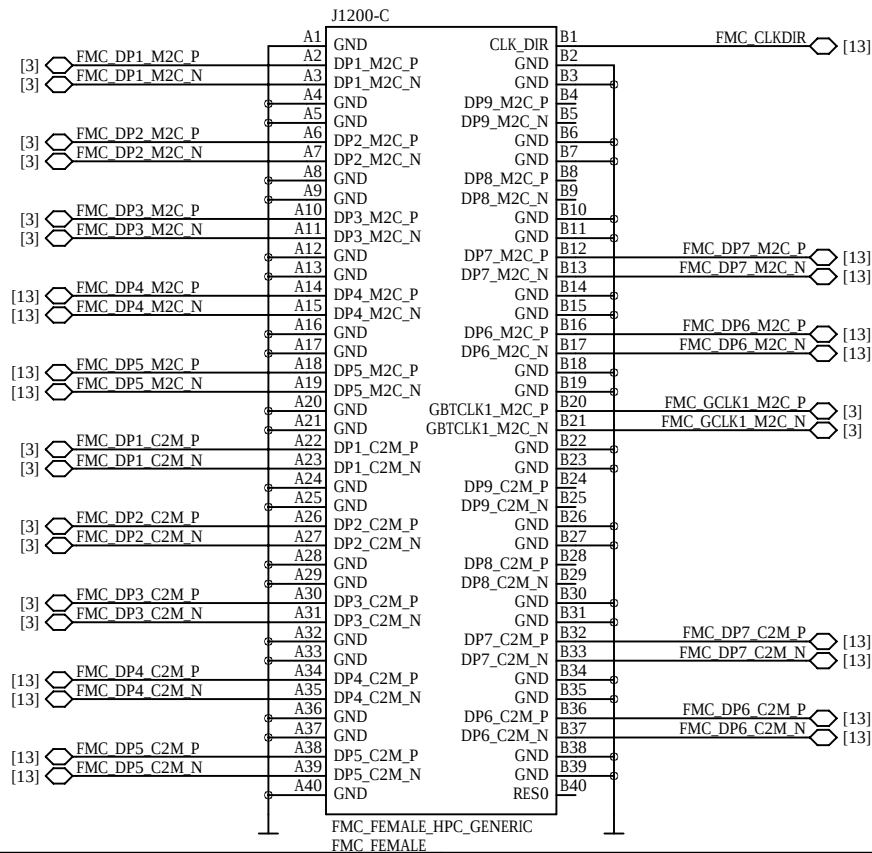
Sheet Name 12_FMC_HPC_CONNECTOR_0
Project Mercury PE1

Customer No 0000
Project No 423
Revision R4.6
Designed MHEI
DNE = do not equip
Date 13 Apr 2021
Confidential
Sheet/sheets 12 / 23



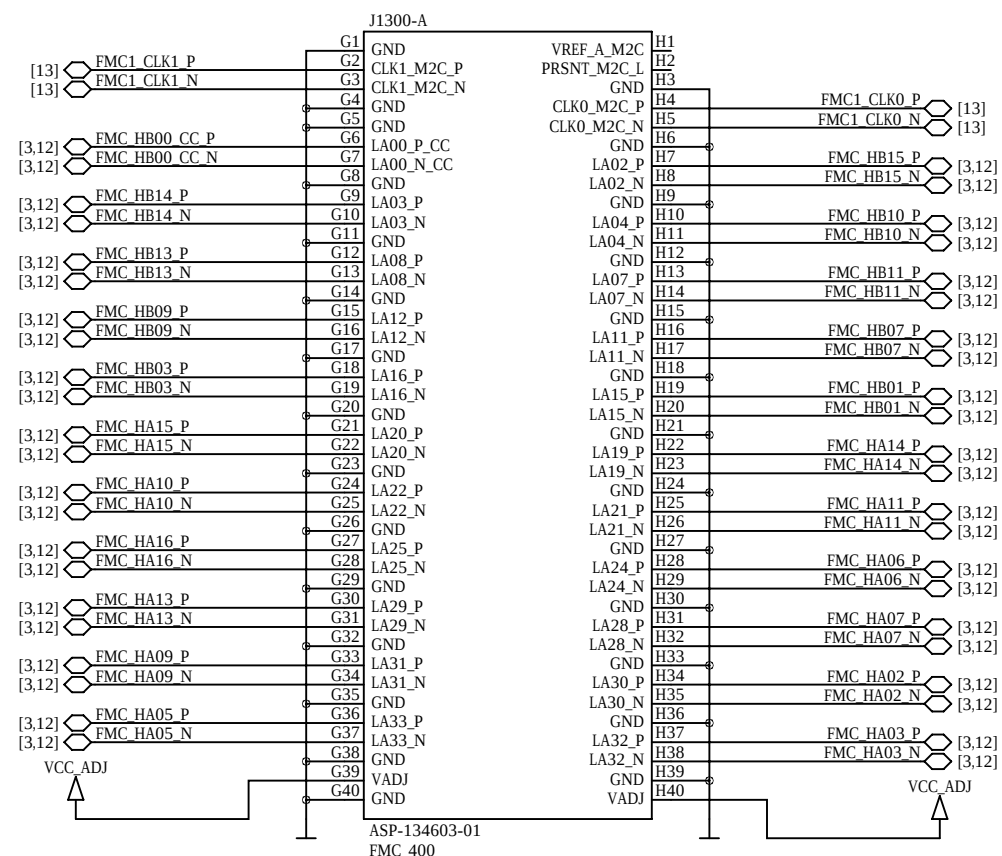
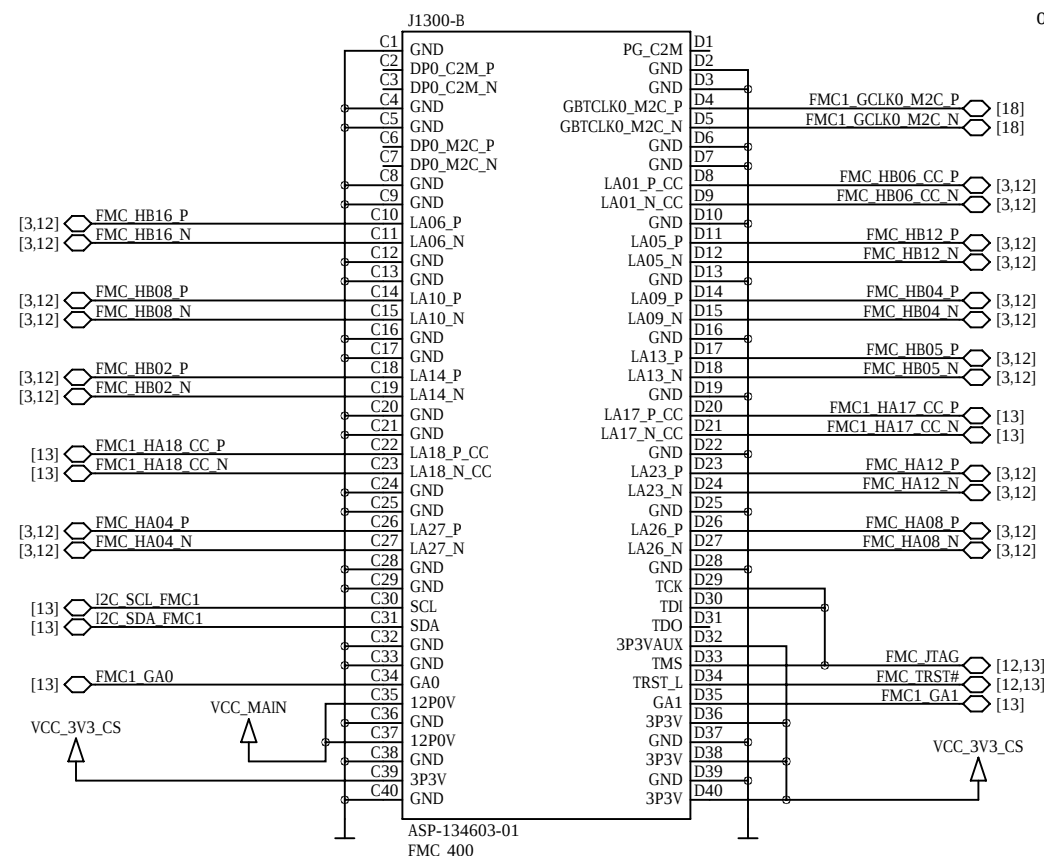
FMC HPC Connector 0

optional in some configurations

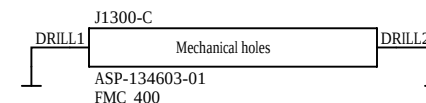
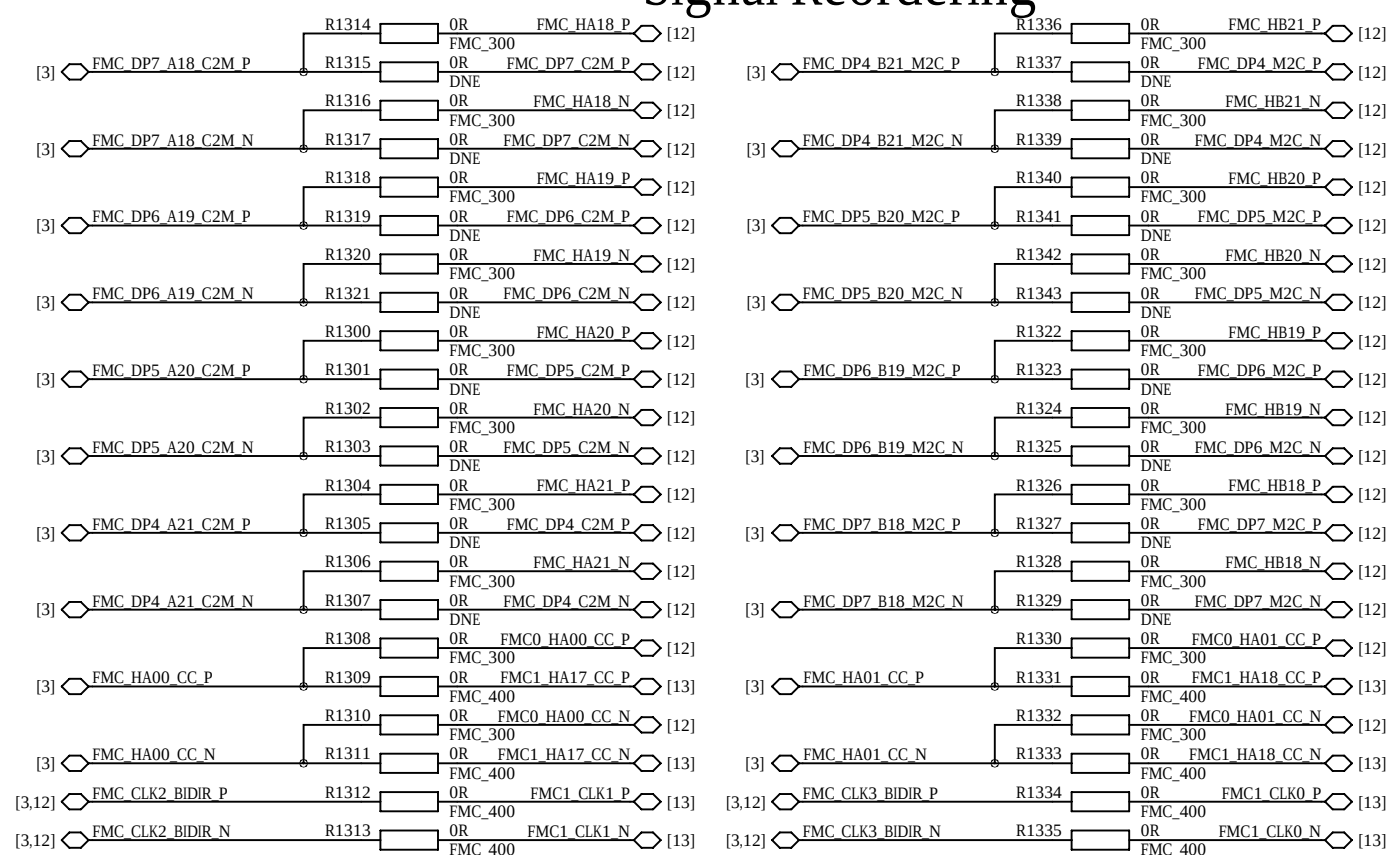


FMC LPC Connector 1

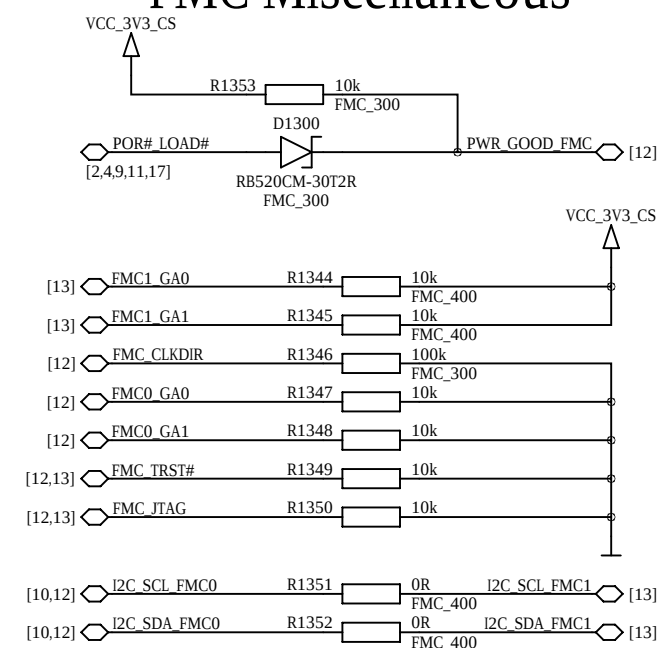
optional in some configurations



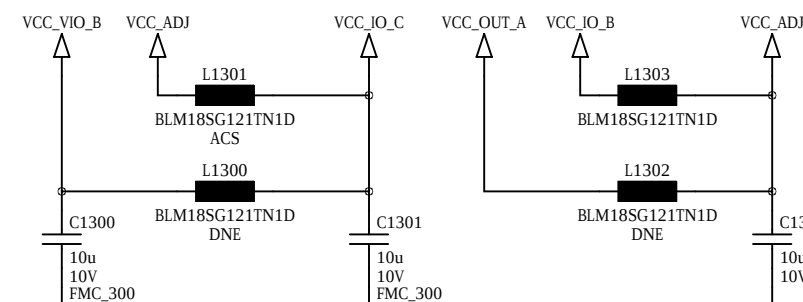
Signal Reordering



FMC Miscellaneous



FMC Power



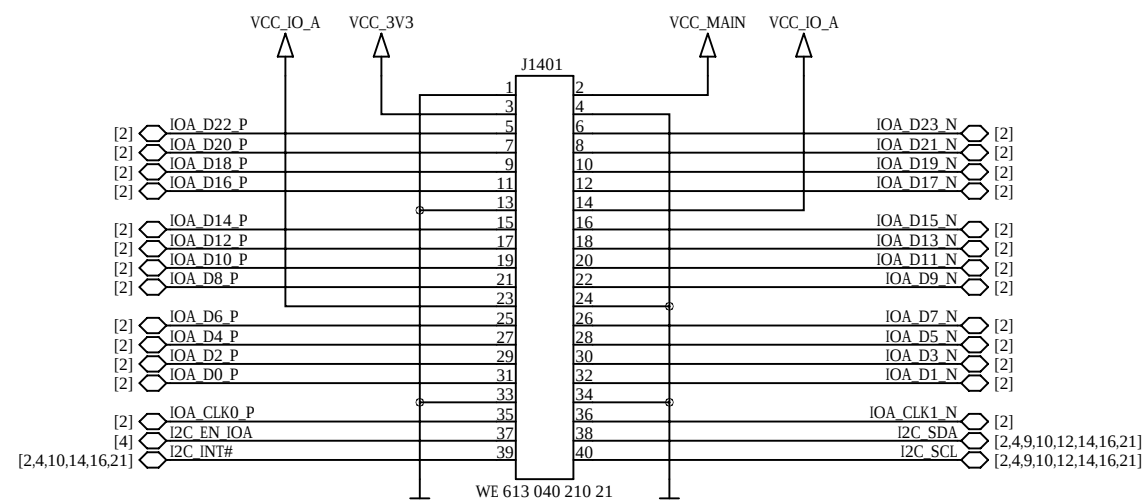
FMC Infos

suggested connector for FMC HPC modules: ASP-134488-01

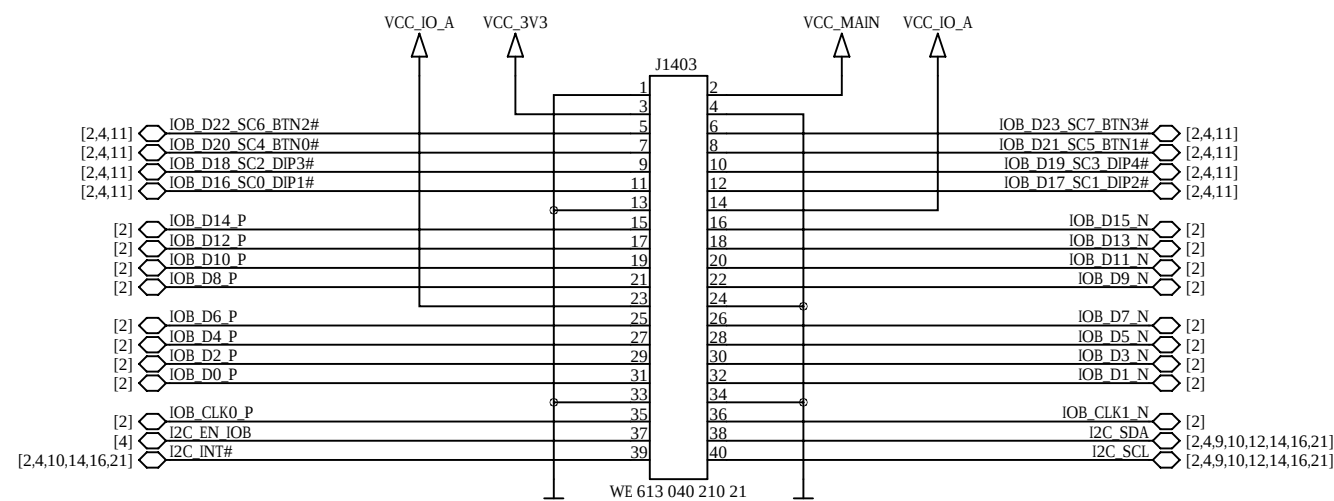
suggested connector for FMC LPC modules: ASP-134604-01



Anios IO Connector A

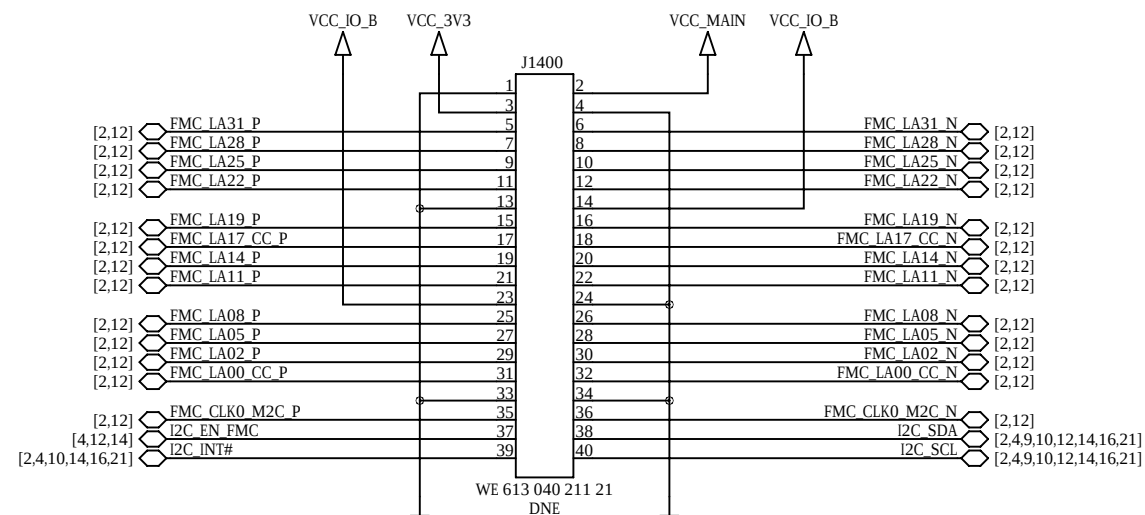


Anios IO Connector B



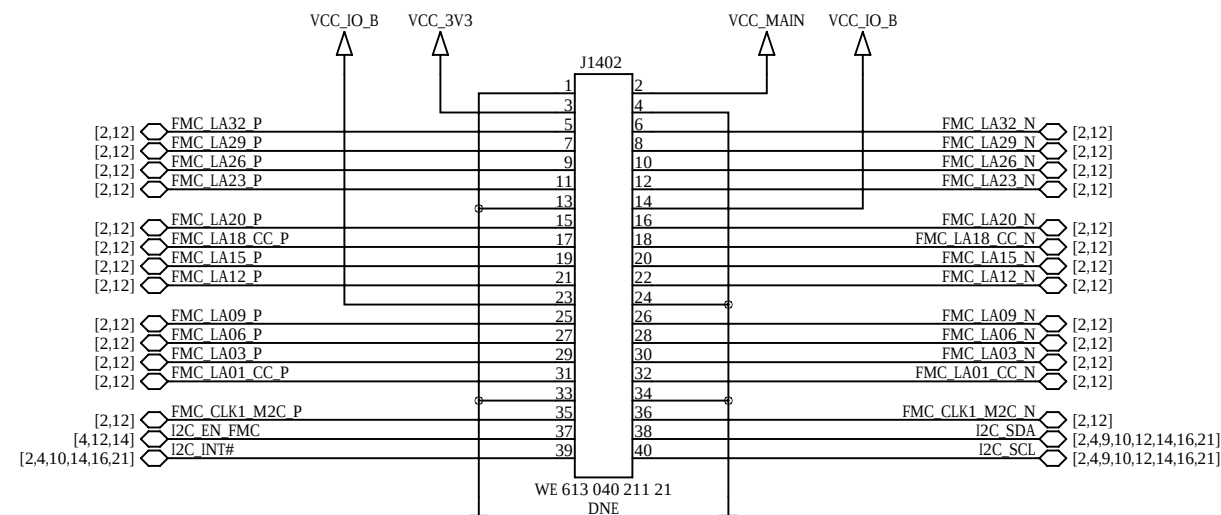
Anios IO Connector F

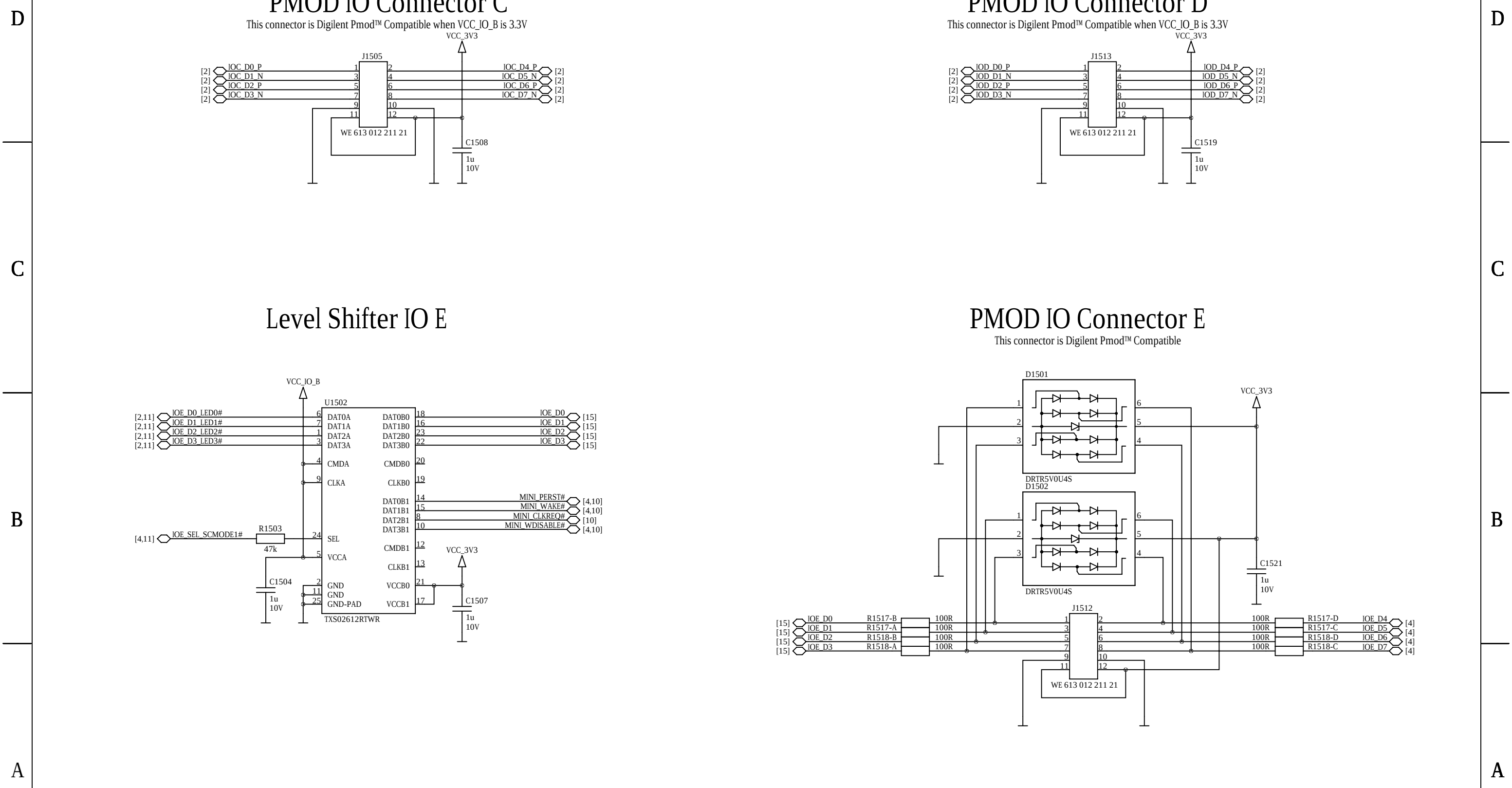
optional, shared with FMC connector



Anios IO Connector G

optional, shared with FMC connector





D

C

B

A

D

C

B

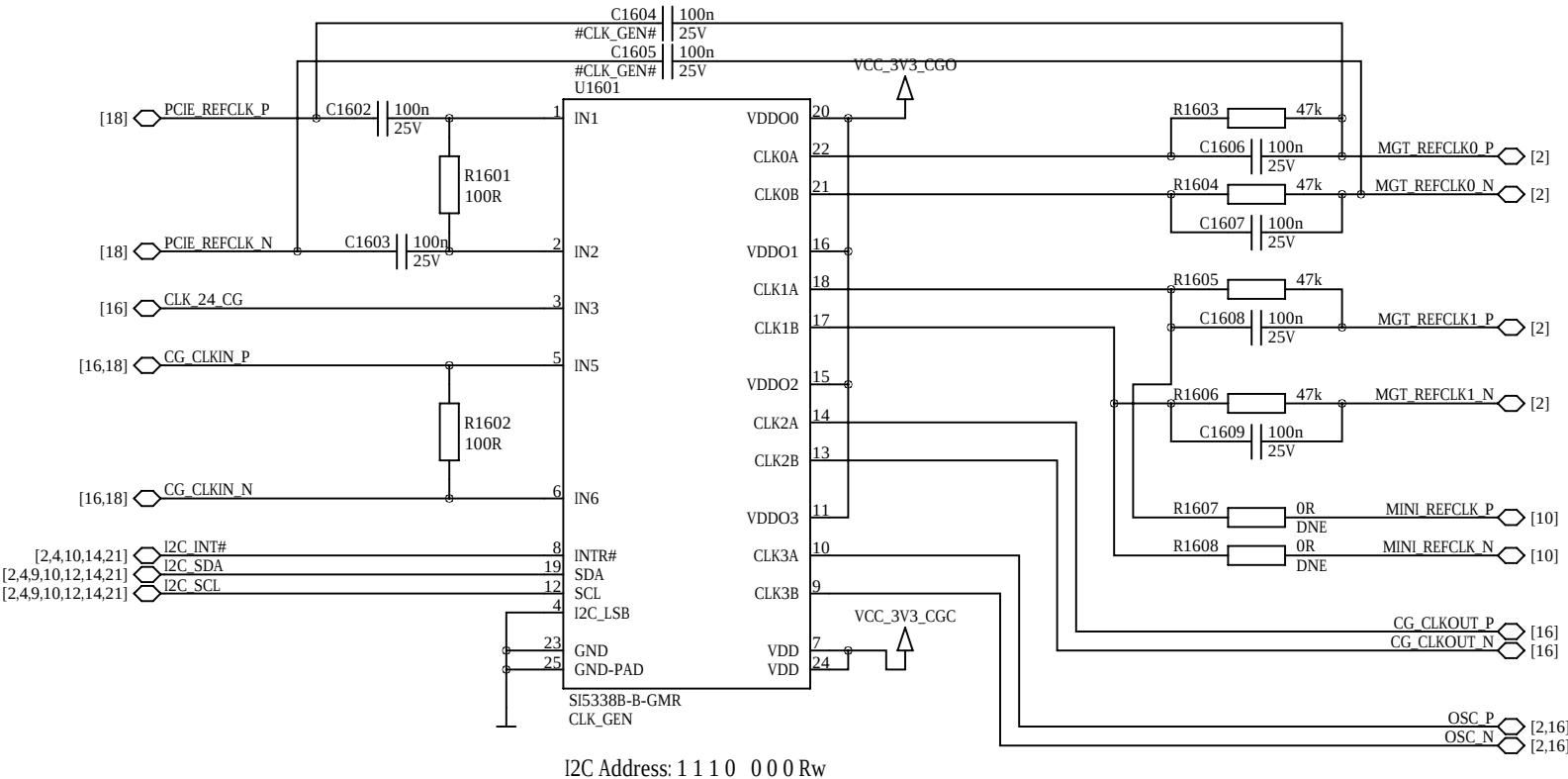
A

Clock Generator

Configuration must be implemented over I2C at startup.

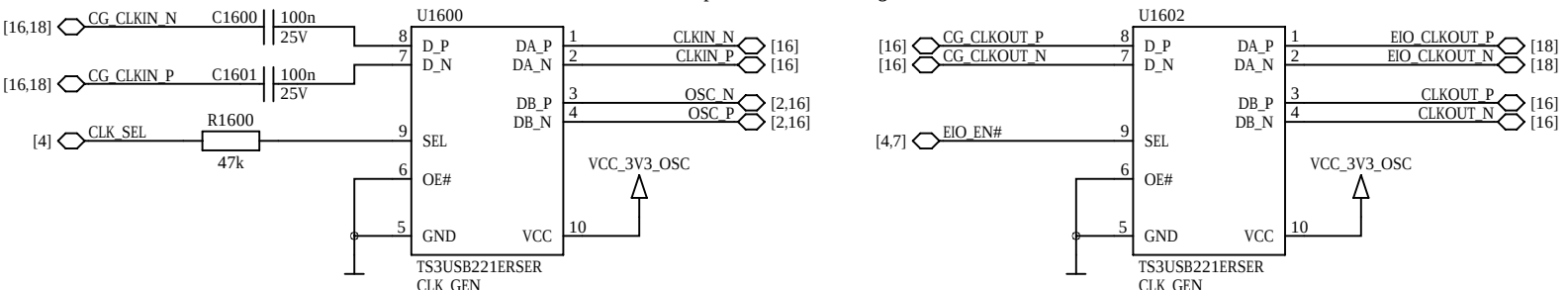
Generates up to four LVDS/HCSL/LVCMOS clocks of up to 350 MHz.

NVM OTP configuration memory can only be written once and doing so voids the board warranty.

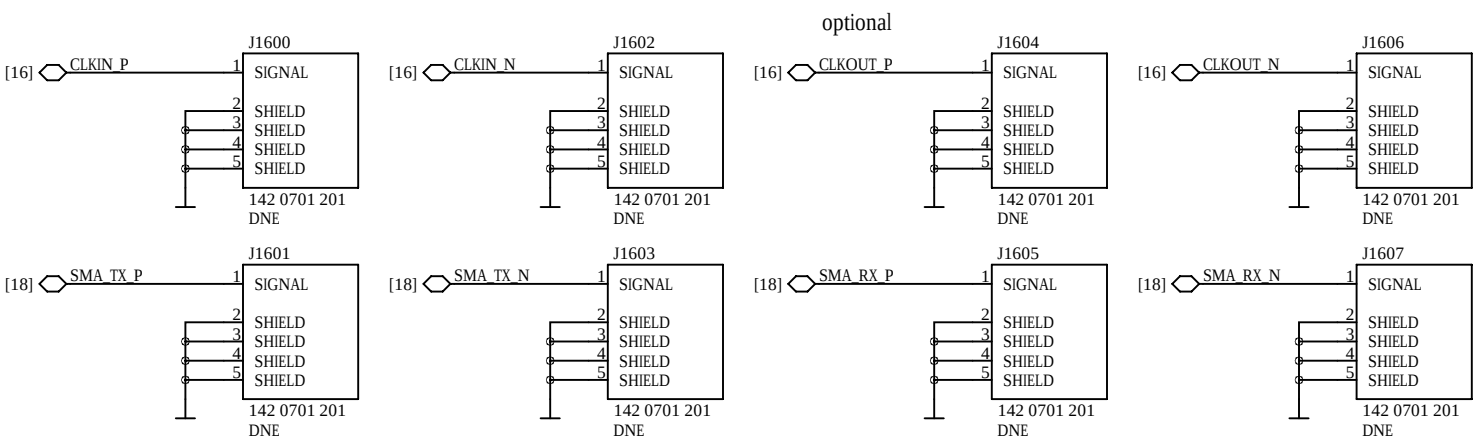


Clock Multiplexers

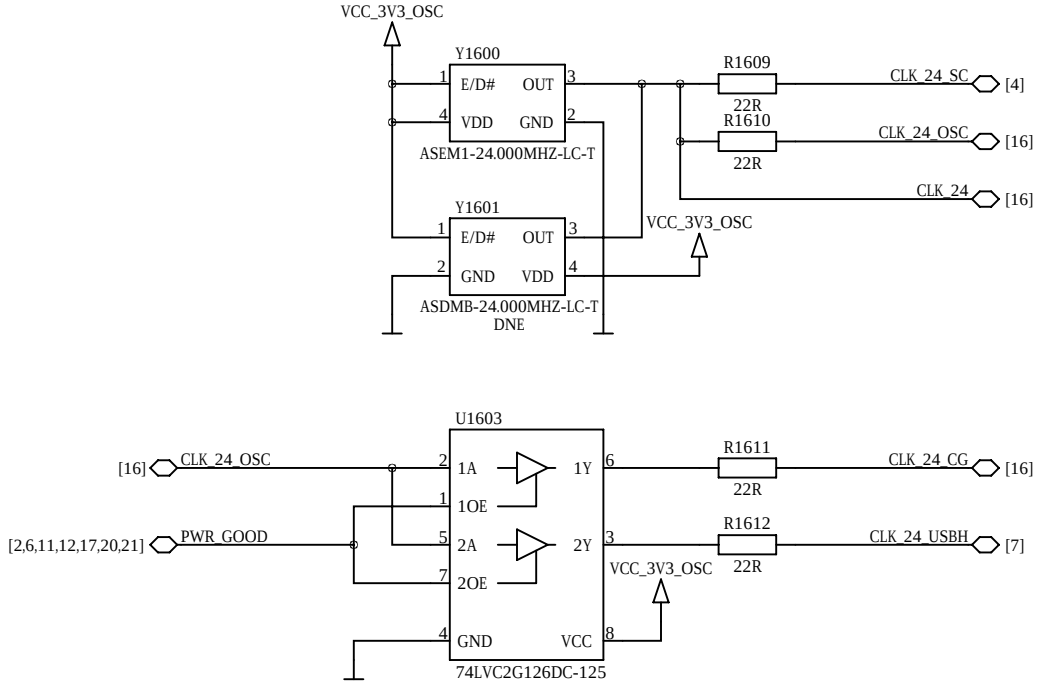
optional in some configurations



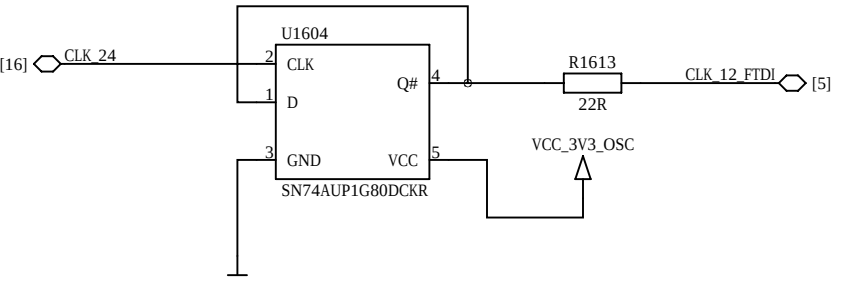
Clock SMA Connectors



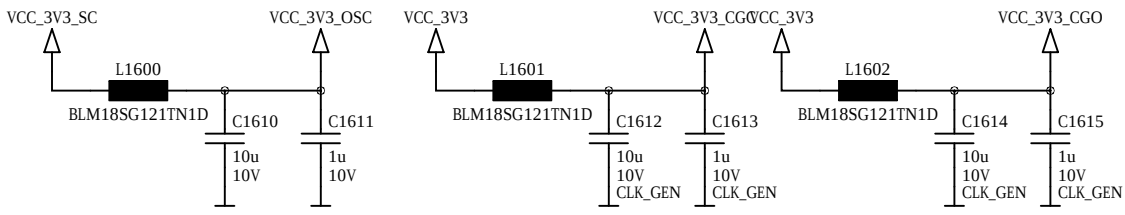
Oscillator



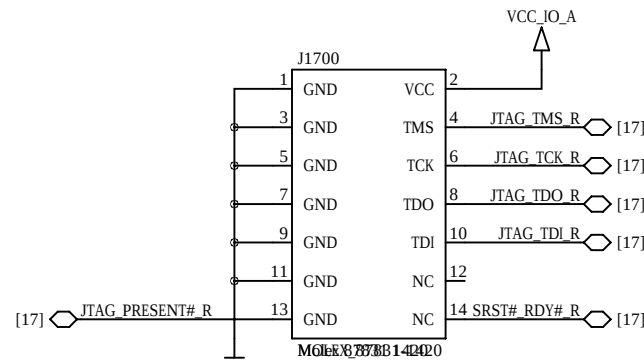
Clock Divider



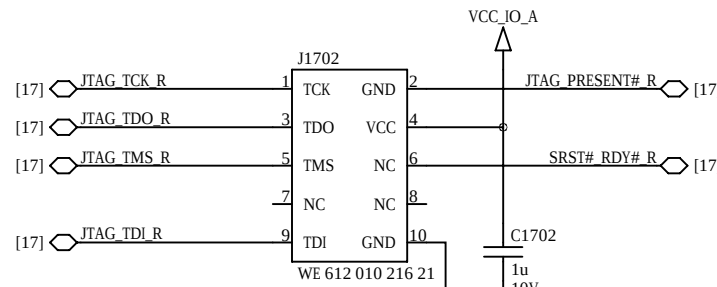
Clock Generator Power Filters



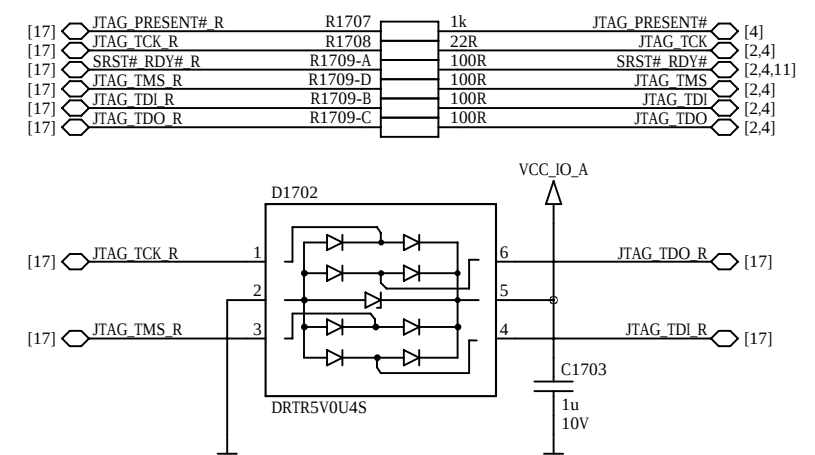
Xilinx JTAG Connector



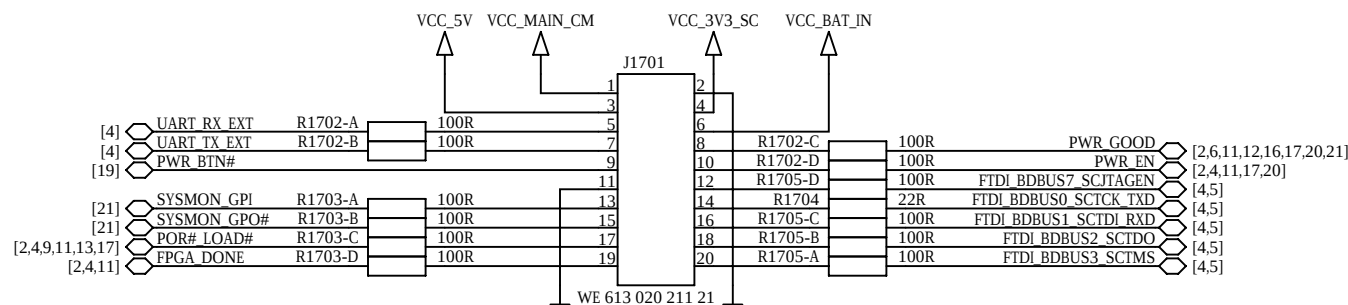
Altera JTAG Connector



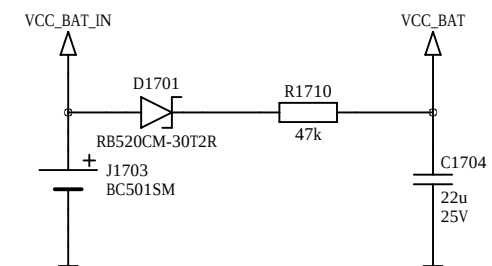
JTAG ESD Protection



Control Connector

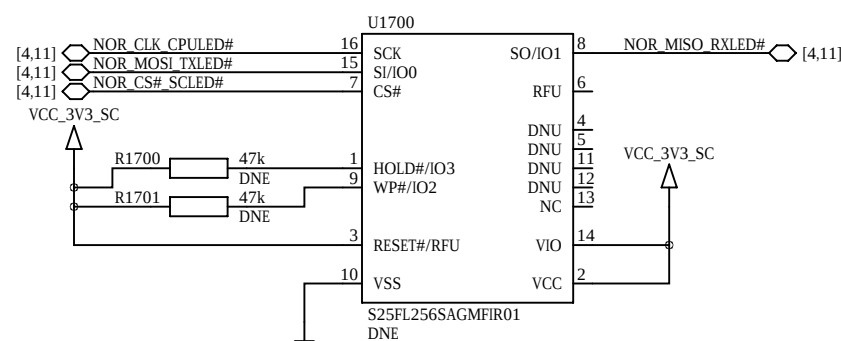


Battery Holder



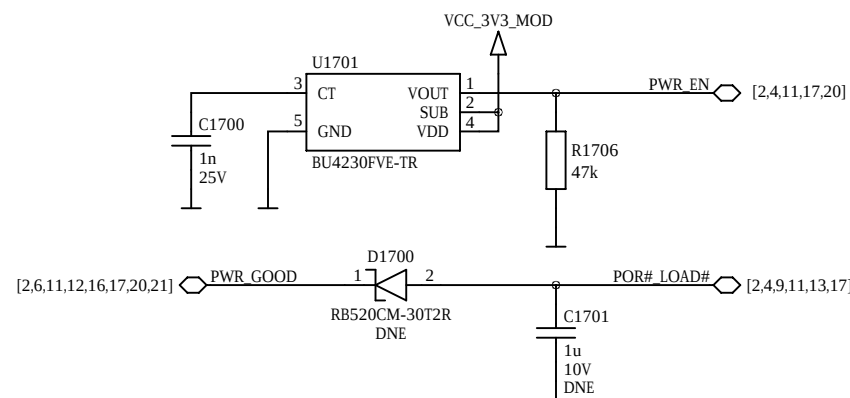
SPI NOR Flash

optional
pin compatible with Altera EPCS/EPCQ and Micron N25Q series



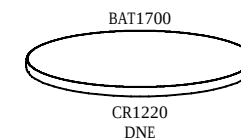
Reset Circuit

D1700, C1701 only needed when not using System Controller

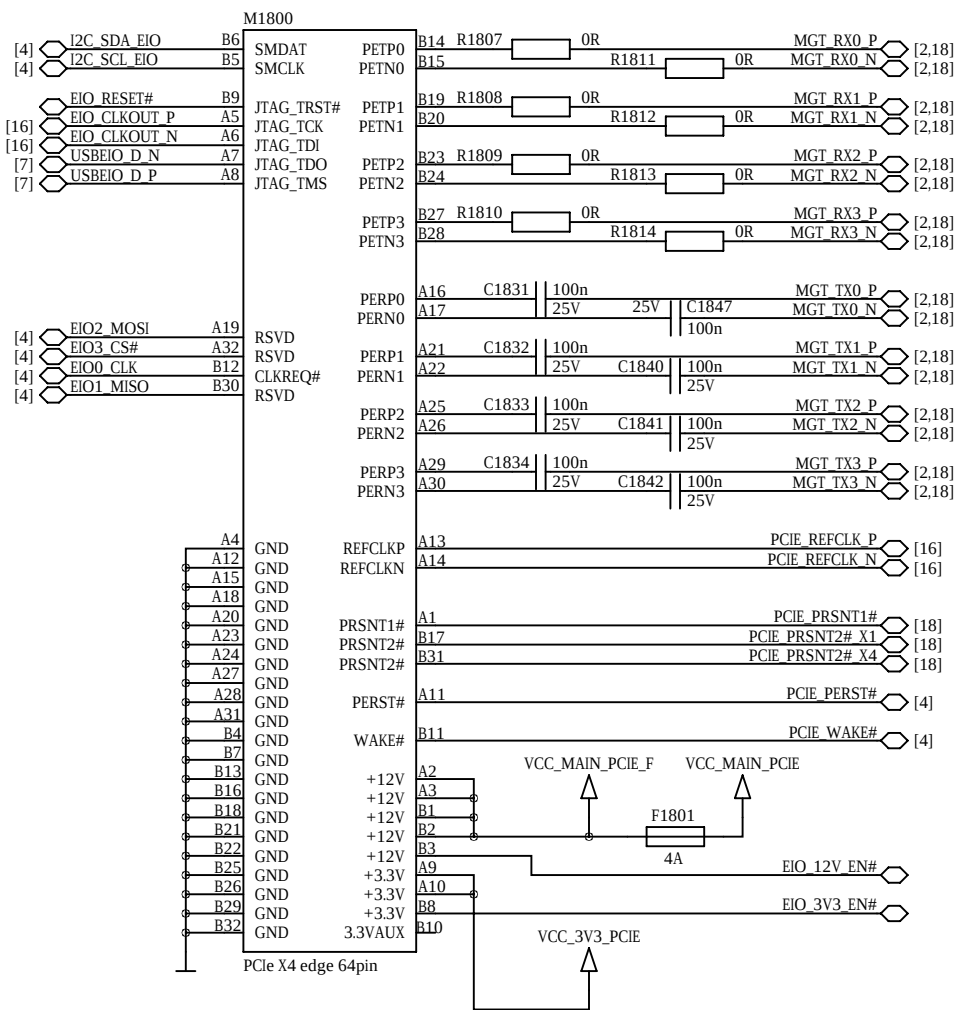


Battery

optional



PCIe x4 Edge Connector

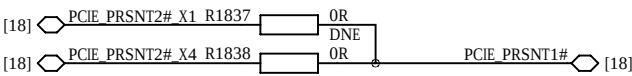


Edge IO Power

not included in user schematics

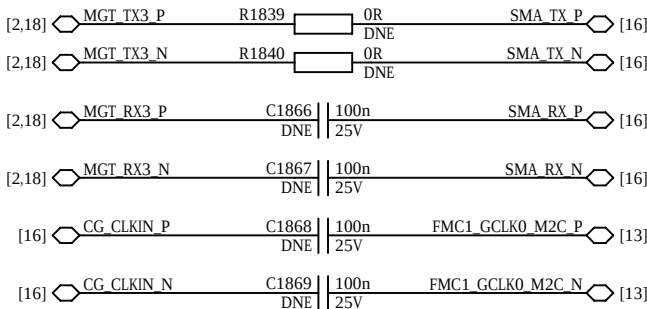
This part is intentionally left blank.

PCIe Present



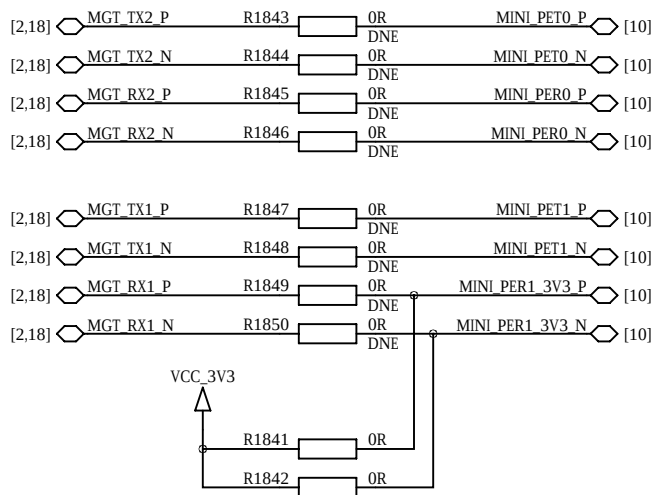
FMC MGT Signals

optional



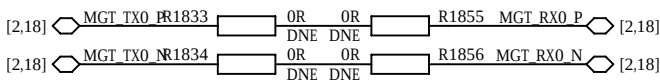
Mini PCIe / Mini SATA Card Signals

optional



MGT Loopback

optional



D

D

C

C

B

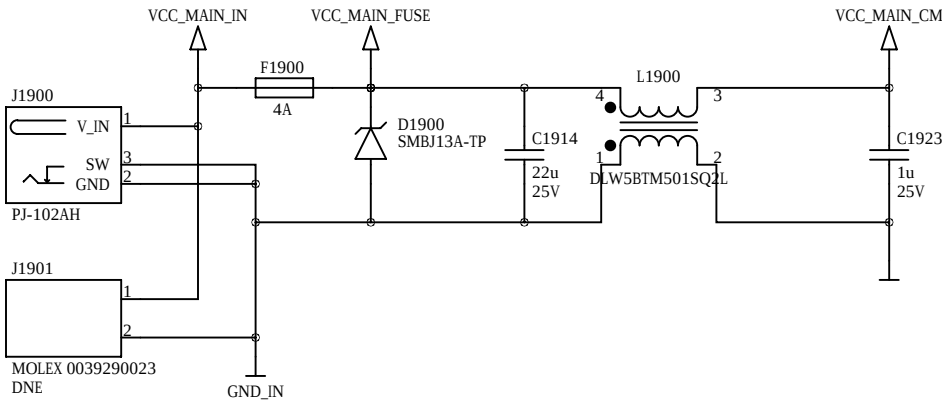
B

A

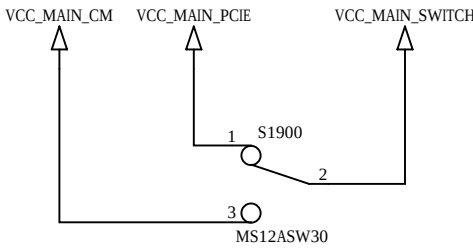
A

External Power Input

12V 2.5A

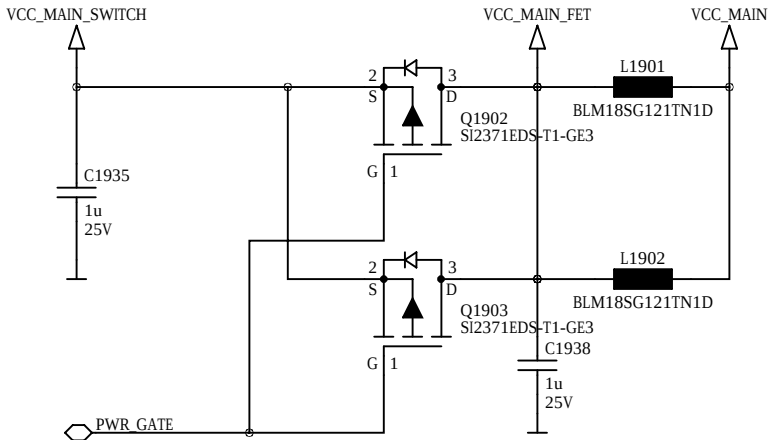


Power Slide Switch



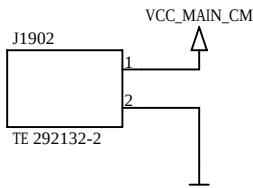
Default: 1-2

Power Input Control

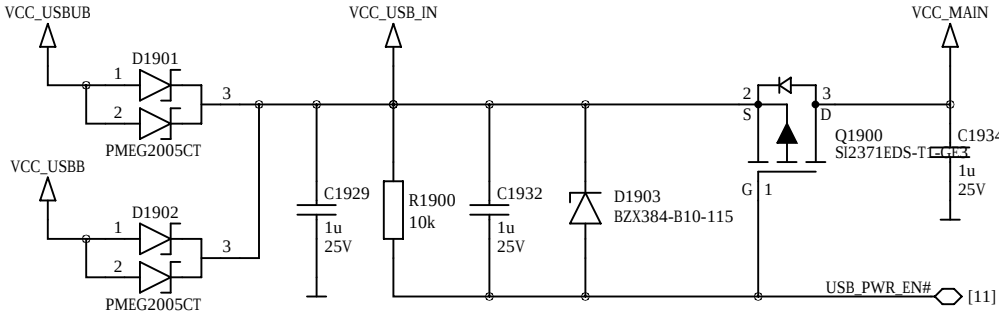


Internal Power Input

12V 4A



USB Power Input Control

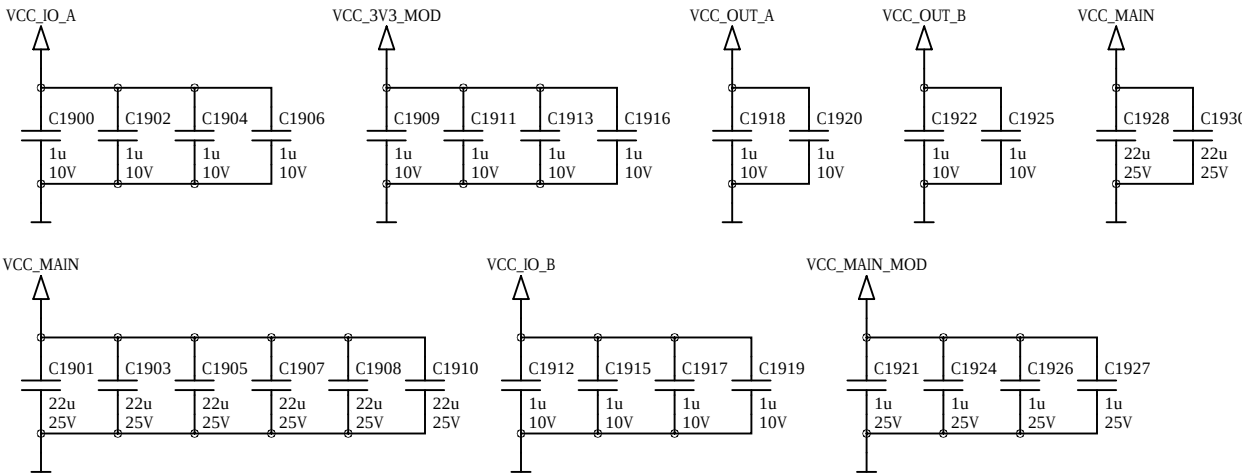


Power Switch Control

not included in user schematics

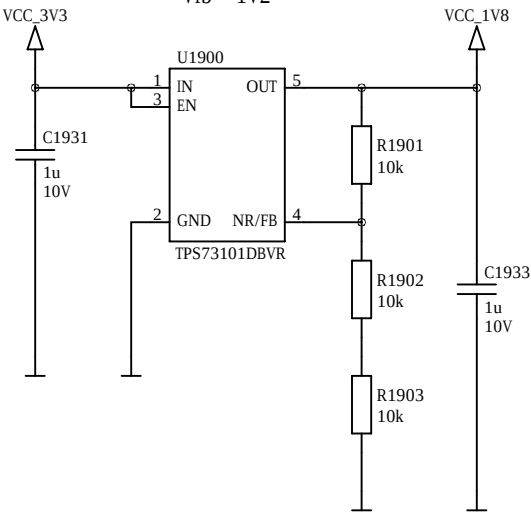
This part is intentionally left blank.

Power Decoupling

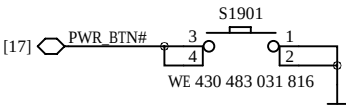


LDO 3V3 to 1V8 0.15A

Vfb = 1V2

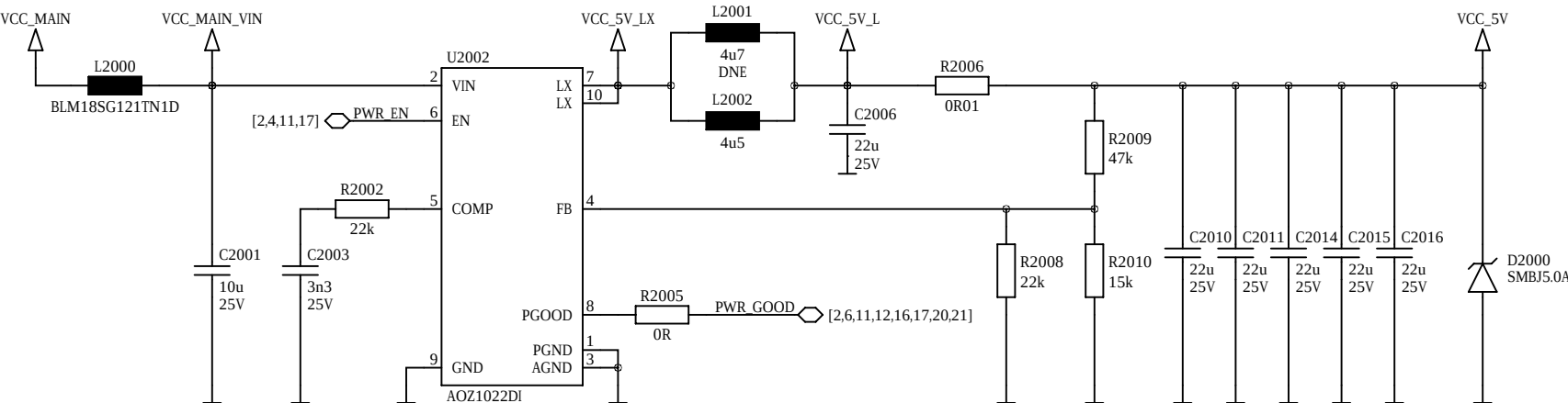


Power Button



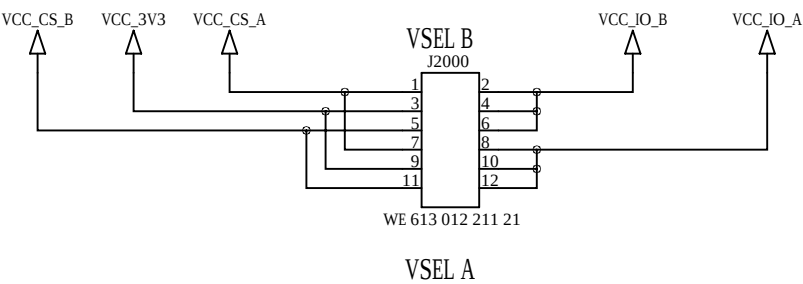
Step-Down Converter 12V to 5V 3A

Vfb = 0.8V fs = 500kHz



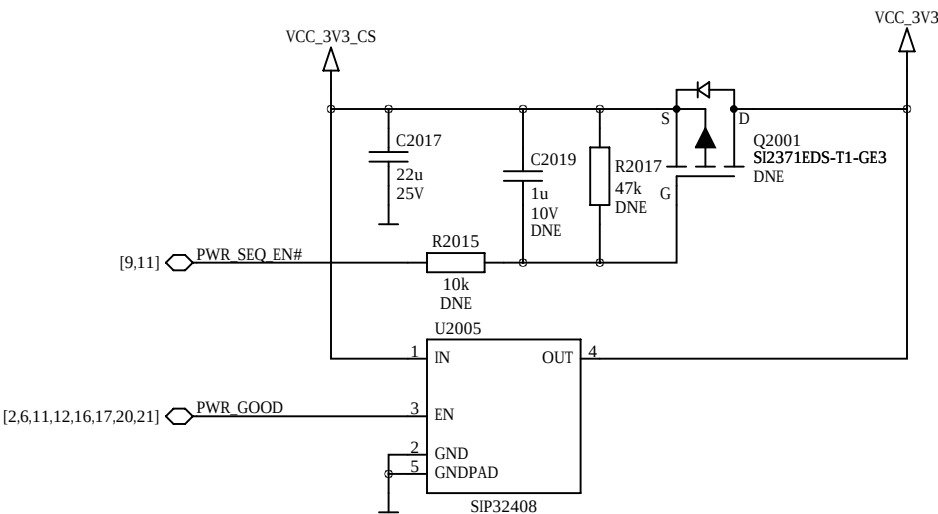
Do not touch feedback and compensation network during operation!
When using power over USB, remove R2005

IO Voltage Select

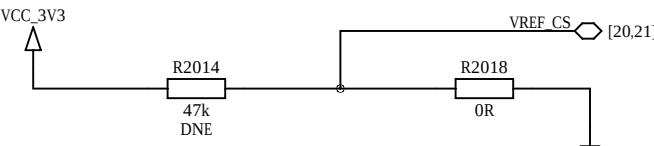


Default Jumpers: 2-4, 8-10
The jumpers need to be set correctly for the users application respecting the modules supported IO voltage ranges.

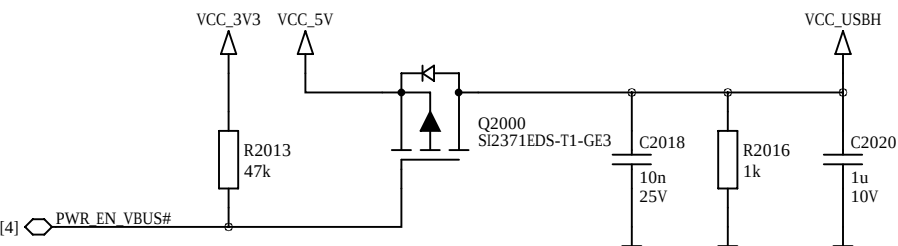
Power Sequencing



Current Sense Reference

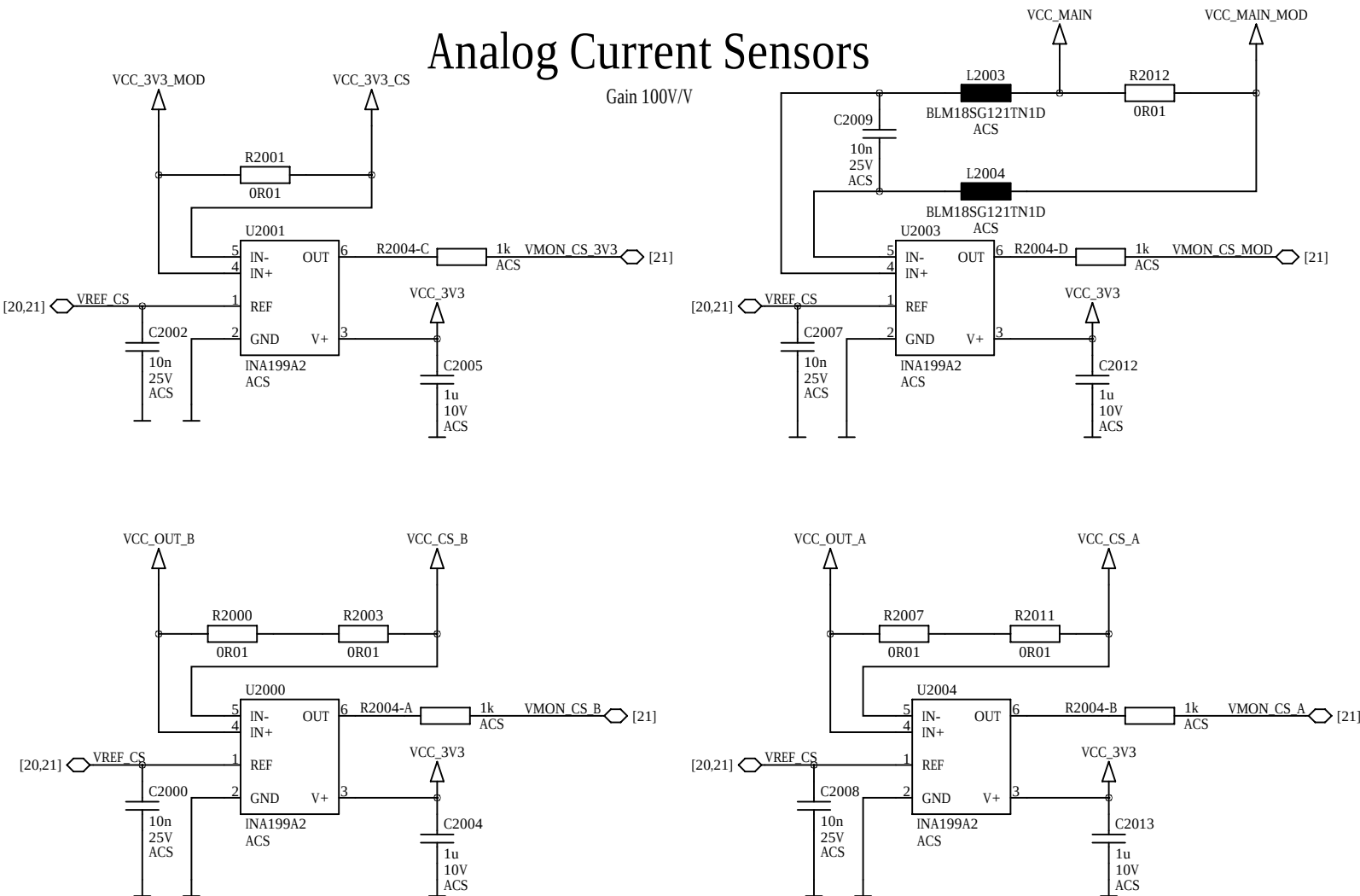


USB VBUS Control

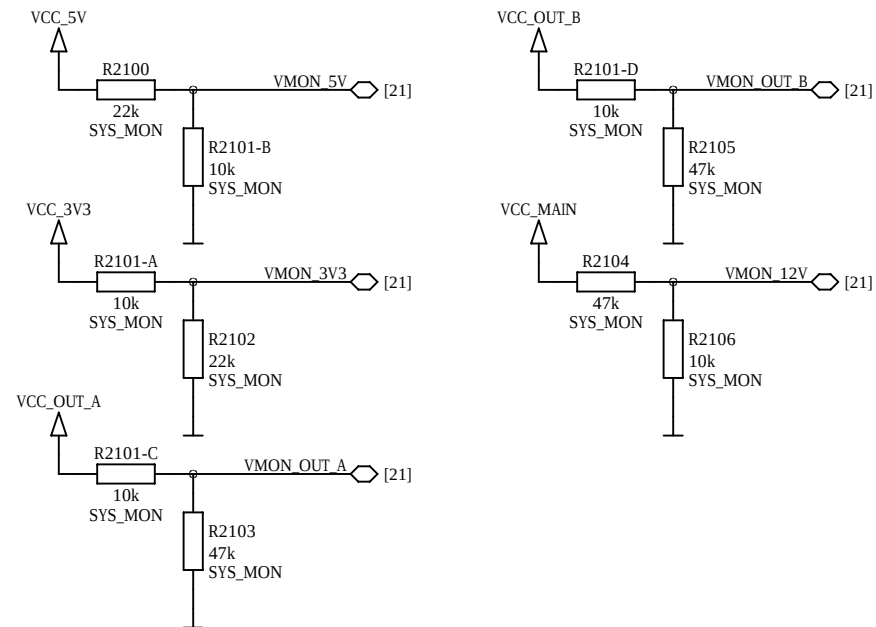


Analog Current Sensors

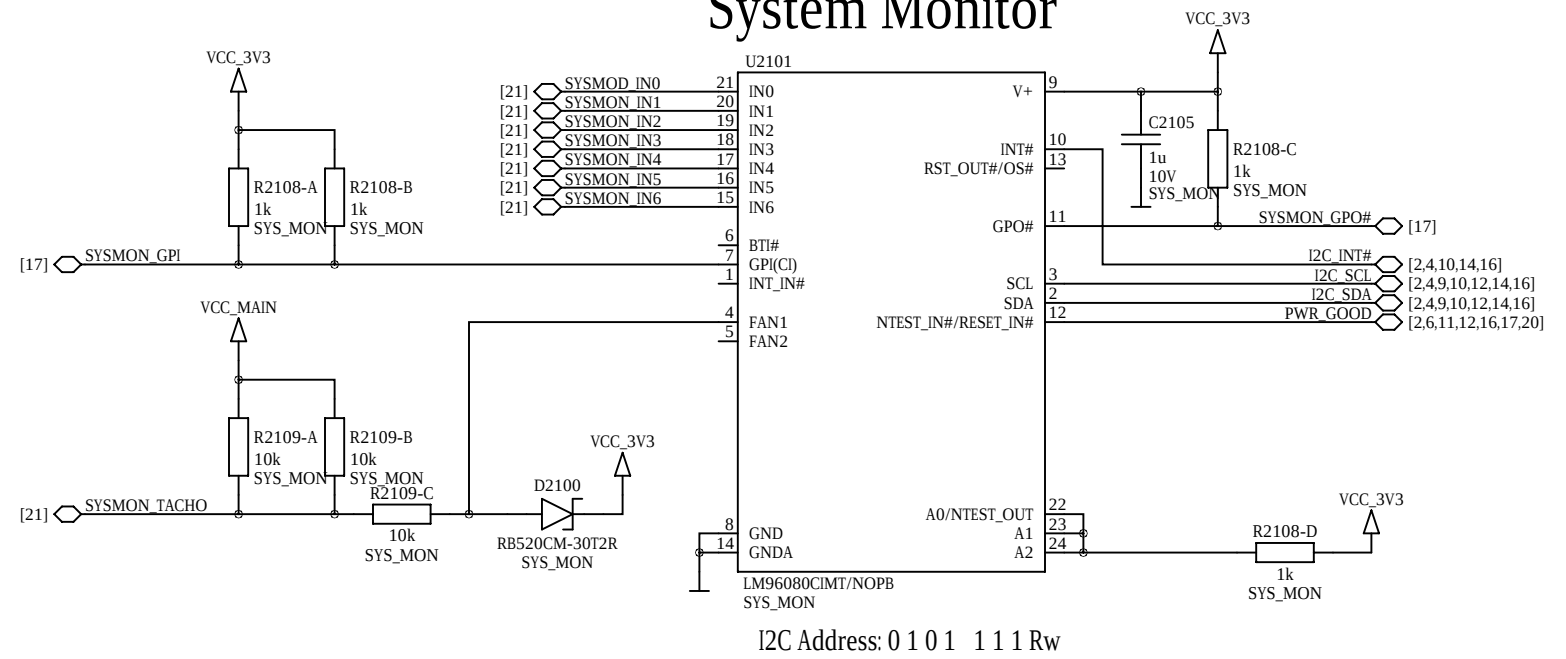
Gain 100V/V



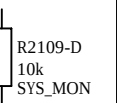
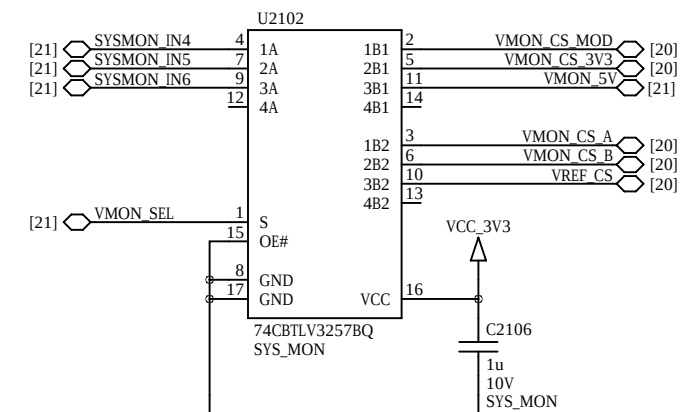
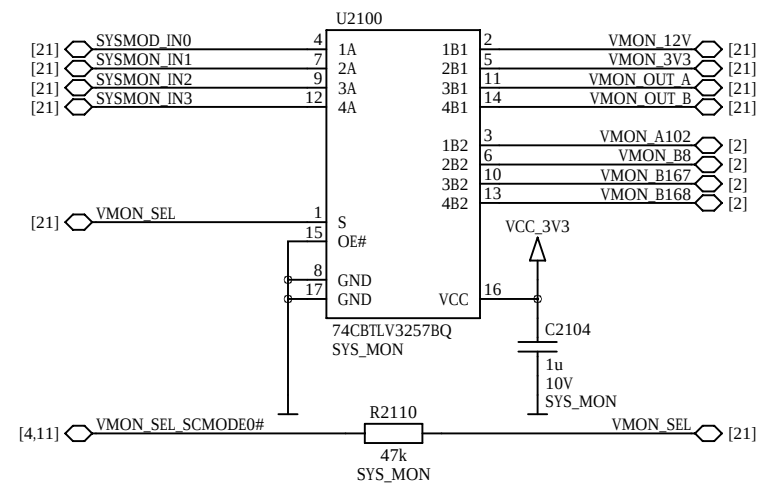
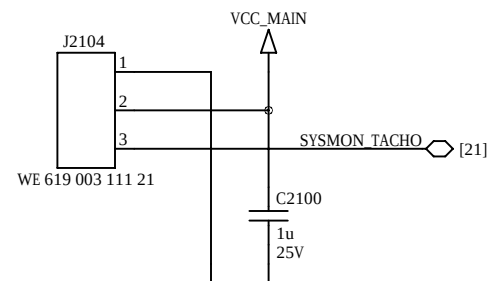
System Monitor Voltage Sense



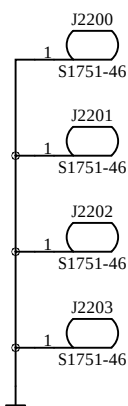
System Monitor



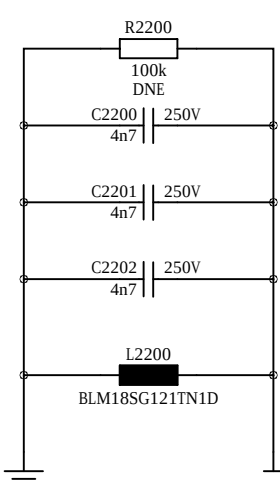
System Monitor Analog Multiplexers



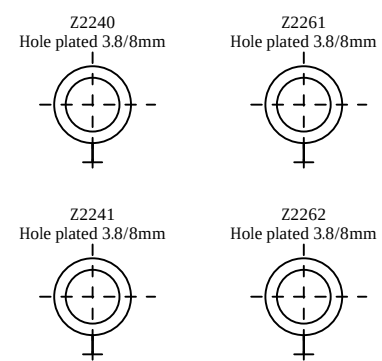
Ground Test Points



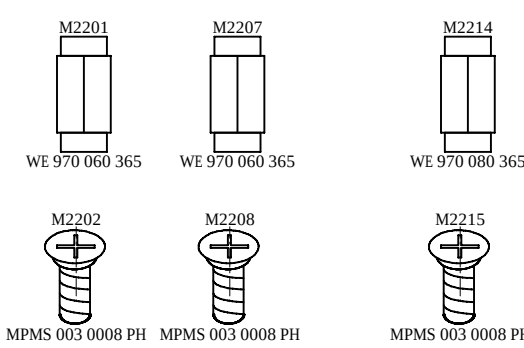
Chassis Ground



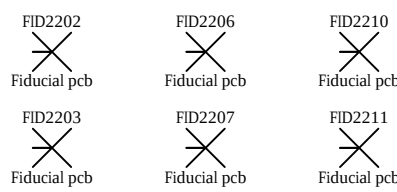
Board Mounting Holes M3



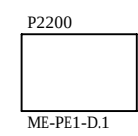
Board Spacers



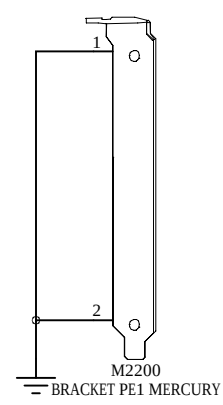
Fiducials



PCB

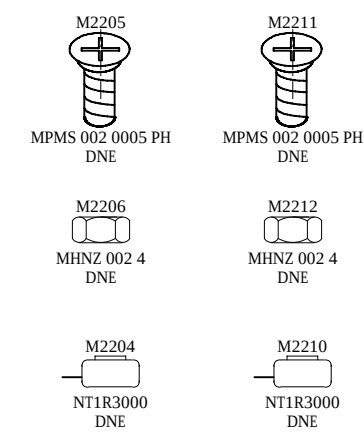


PCIe Bracket

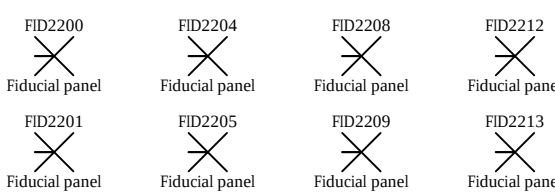


Mini Card Mounting

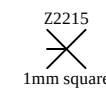
optional



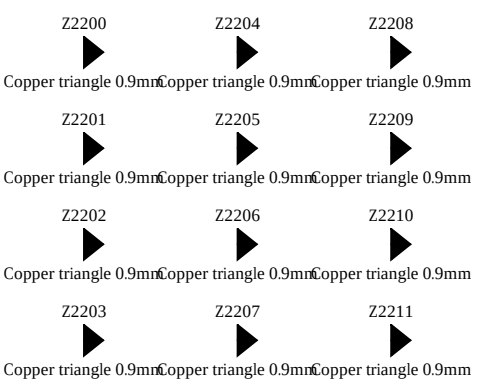
Panel Fiducials



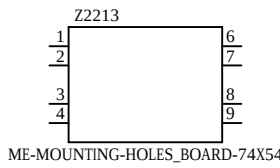
Module Pin-1 Marking



Connector Pin-1 Marking



Module Mounting Holes M3



Text

Z2224 Text USB 12	Z2231 Text USB 34	Z2238 Text CTRL	Z2247 Text PWGD	Z2254 Text DONE	Z2263 Text RX	Z2270 Text TX	Z2277 Text CPU	Z2284 Text SC
Z2225 Text MINI	Z2232 Text FAIL	Z2239 Text CFG A	Z2248 Text CFG B	Z2255 Text USER	Z2264 Text JTAG	Z2271 Text SD	Z2278 Text USBH	Z2285 Text USB3
Z2218 Text IO A	Z2226 Text IO B	Z2233 Text IO C	Z2242 Text IO D	Z2249 Text IO E	Z2256 Text IO F	Z2265 Text IO G	Z2272 Text ETH0	Z2279 Text ETH1
Z2219 Text CLKIN-	Z2227 Text CLKIN+	Z2234 Text CLKOUT-	Z2243 Text CLKOUT+	Z2250 Text FMC	Z2257 Text LPC	Z2266 Text PCIE	Z2273 Text 12VDC	Z2280 Text USBUB
Z2220 Text DCIN	Z2228 Text FAN	Z2235 Text PWR	Z2244 Text VSEL A	Z2251 Text VSEL B	Z2258 Text POR	Z2267 Text SRST	Z2274 Text A	Z2281 Text B
Z2221 Text 3V3	Z2229 Text 1	Z2236 Text 2	Z2245 Text 3	Z2252 Text 4	Z2259 Text A	Z2268 Text B	Z2275 Text 3V3	Z2282 Text 0
Z2222 Text 1	Z2230 Text 2	Z2237 Text 3	Z2246 Text Mercury PE1	Z2253 Text ON	Z2260 Text B	Z2269 Text RX+	Z2276 Text RX-	Z2283 Text TX+
Z2223 Text TX-								

Component Variants

SYS_MON	FMC	EMMC	SIMCARD	MOD_CONN	ACS	CLK_GEN	FMC_300	FMC_400
page 21	Part J1200A ASP-134603-01 EN100364 FMC_HPC_LPC_GENERIC Part J1200B ASP-134486-01 EN101301 FMC_HPC_LPC_GENERIC	Part U900A:1 EMMC04G-M627-X03U EN102717 Part U900A:2 H26M31001HPR EN101269	page 10	page 3	pages 6, 7, 13 and 20	page 16	page 13	page 13

Assembly Variants

Assembly Variant	SYS_MON	FMC	EMMC	SIMCARD	MOD_CONN	ACS	CLK_GEN	FMC_300	FMC_400
Variant EN103080:0 ME-PE1-200-C-R4.6	Option EN103080:1 SYS_MON DNE	Option EN103080:2 FMC_FEMALE EN100364	Option EN103080:3 EMMC DNE	Option EN103080:4 SIMCARD DNE	Option EN103080:5 MOD_CONN DNE	Option EN103080:6 ACS DNE	Option EN103080:7 CLK_GEN DNE	Option EN103080:8 FMC_300 DNE	Option EN103080:9 FMC_400 DNE
Variant EN103081:0 ME-PE1-300-W-R4.6	Option EN103081:1 SYS_MON	Option EN103081:2 FMC_FEMALE EN101301	Option EN103081:3 EMMC EN101269	Option EN103081:4 SIMCARD	Option EN103081:5 MOD_CONN	Option EN103081:6 ACS	Option EN103081:7 CLK_GEN	Option EN103081:8 FMC_300	Option EN103081:9 FMC_400 DNE
Variant EN103082:0 ME-PE1-400-W-R4.6	Option EN103082:1 SYS_MON	Option EN103082:2 FMC_FEMALE EN100364	Option EN103082:3 EMMC EN101269	Option EN103082:4 SIMCARD	Option EN103082:5 MOD_CONN	Option EN103082:6 ACS	Option EN103082:7 CLK_GEN	Option EN103082:8 FMC_300 DNE	Option EN103082:9 FMC_400